

Integral formulations and numerical methods for a class of half-space interface problems

Manas Rachh

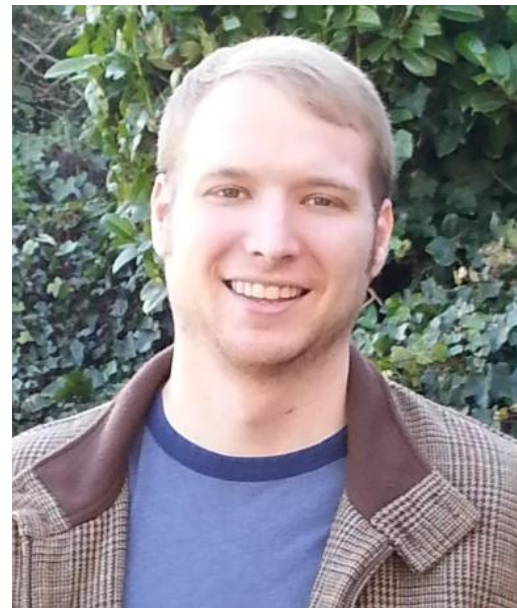
In collaboration with



Peter Nekrasov



Tristan Goodwill



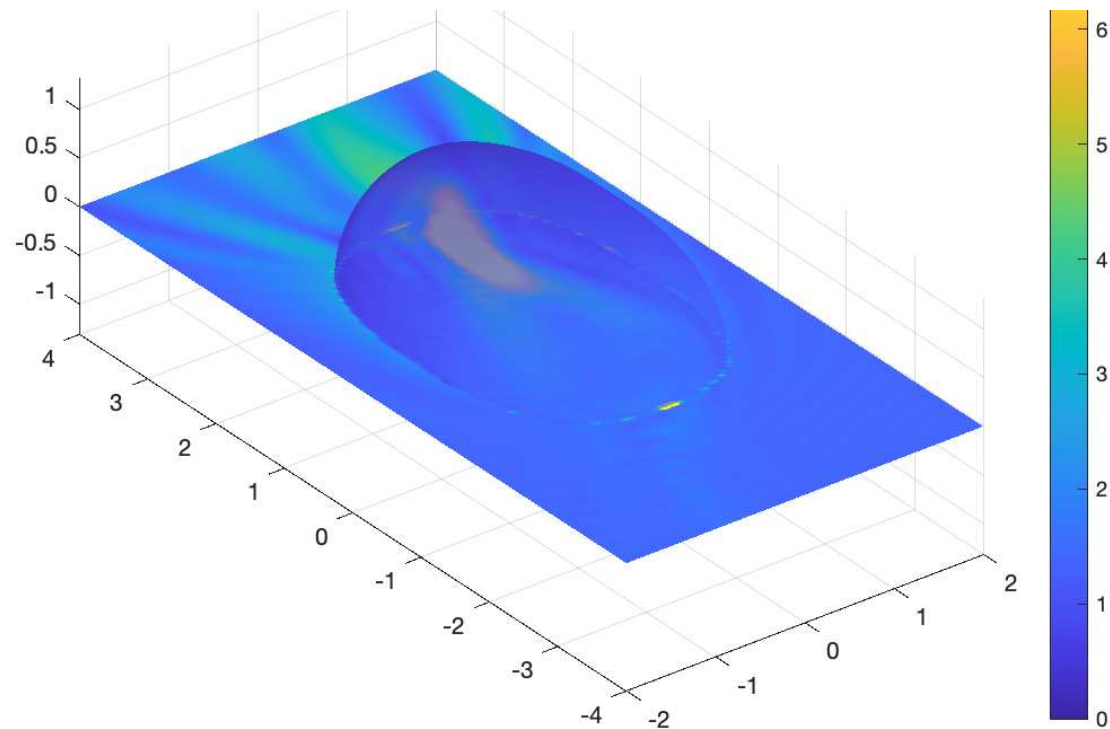
Travis Askham



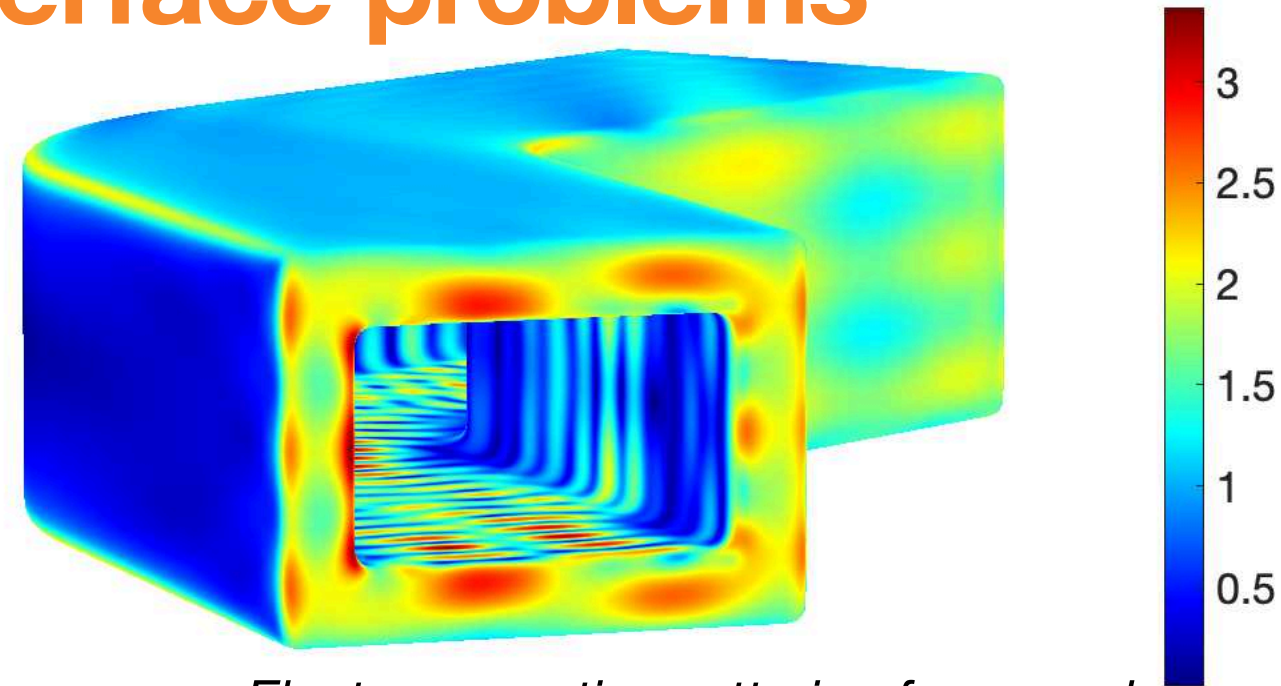
Jeremy Hoskins

- “Integral equations for flexural-gravity waves: analysis and numerical methods” <https://arxiv.org/abs/2501.00887>
- “Surface layers and linearized waves: a boundary integral equation framework” upcoming

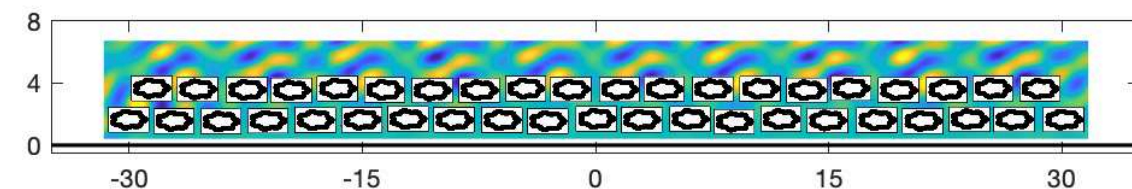
Integral equations and interface problems



Phototaxis in cyanobacteria



Electromagnetic scattering from nearly resonant cavities



Dark matter measurements with antenna arrays

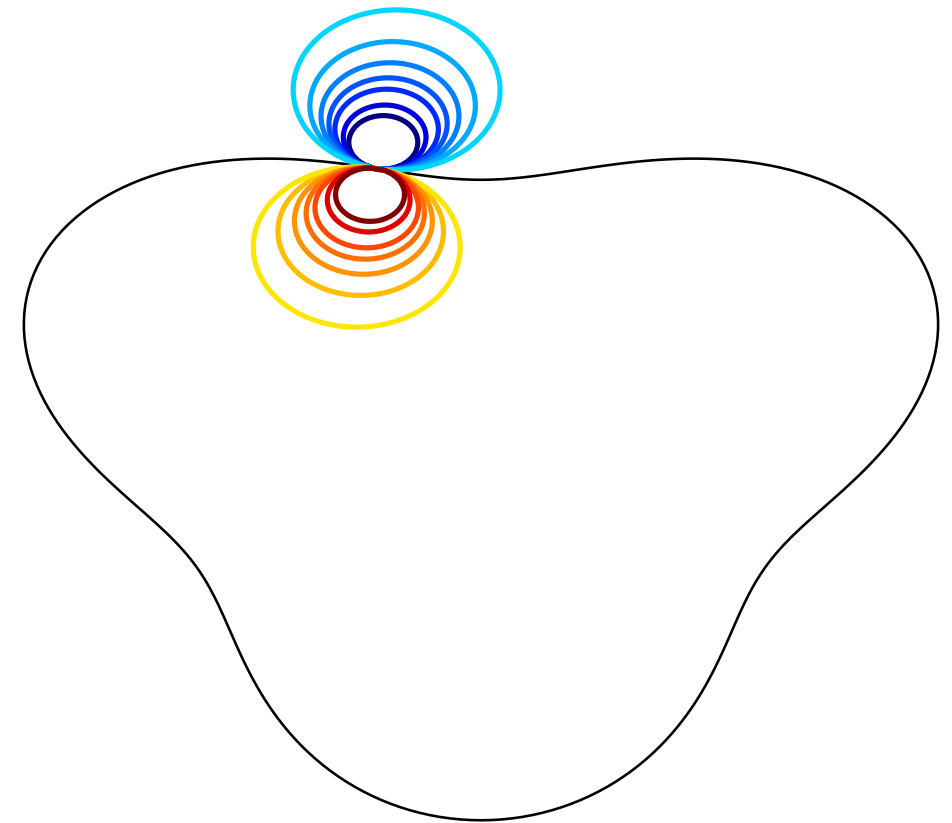
Geophysical modeling of ice-floes

Integral equation methods

$$\begin{aligned}\Delta u &= 0, & x &\in \Omega, \\ u &= f, & x &\in \partial\Omega.\end{aligned}$$

- Analytical ingredients needed: a free space Green's function and a representation/ansatz
- $G(x, y) = -\log |x - y| / (2\pi)$
- Represent u in the form

$$\begin{aligned}u(x) &= \int_{\partial\Omega} \sigma(y) \nabla_y G(x, y) \cdot n(y) ds_y \\ &= \Re \frac{1}{2\pi} \int_{\partial\Omega} \frac{\sigma(z)}{z - \zeta} d\zeta \quad x \in \Omega\end{aligned}$$



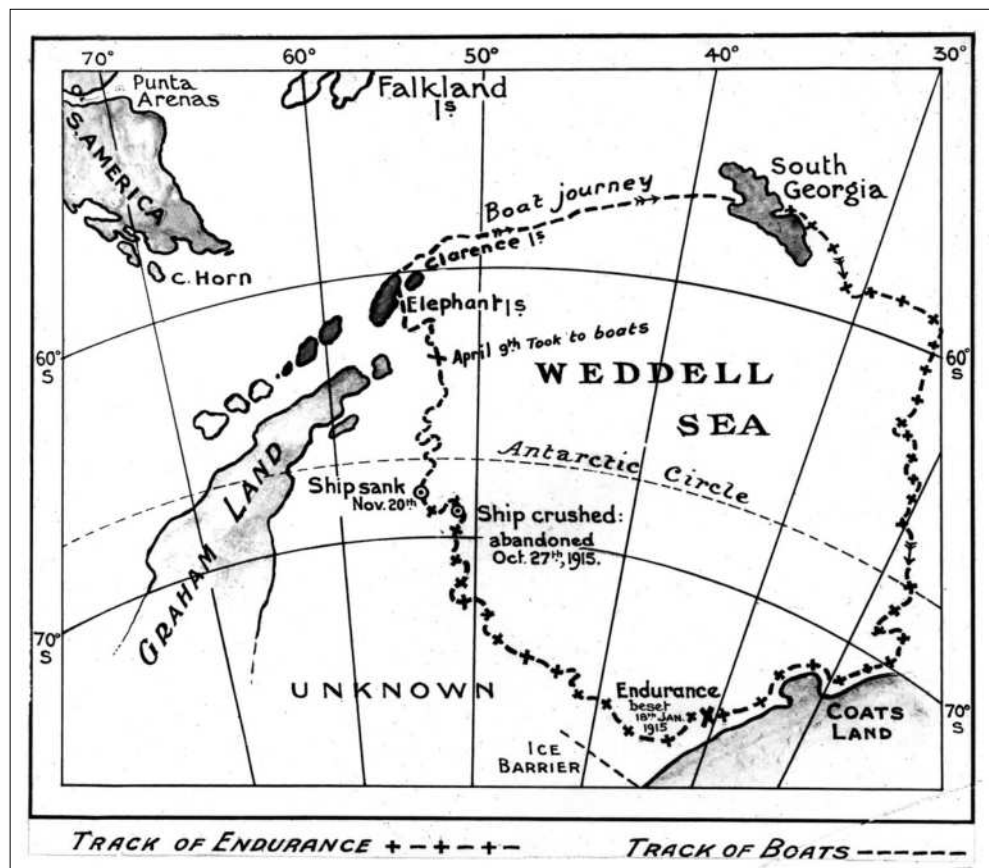
- Taking the limit and using Cauchy's residue theorem

$$-\frac{1}{2}\sigma(x) + \int_{\partial\Omega} \sigma(y) \nabla_y G(x, y) \cdot n(y) ds_y = f(x)$$

The Age of Exploration

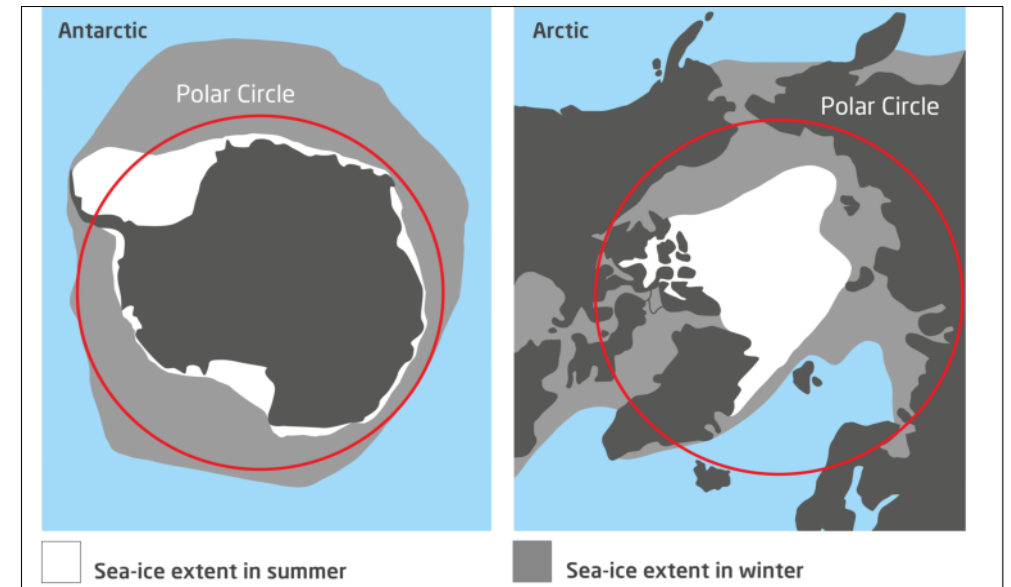
"The noise resembles the roar of heavy, distant surf. Standing on the stirring ice one can imagine it is disturbed by the breathing and tossing mighty giant below."

- Ernest Shackleton, 1919

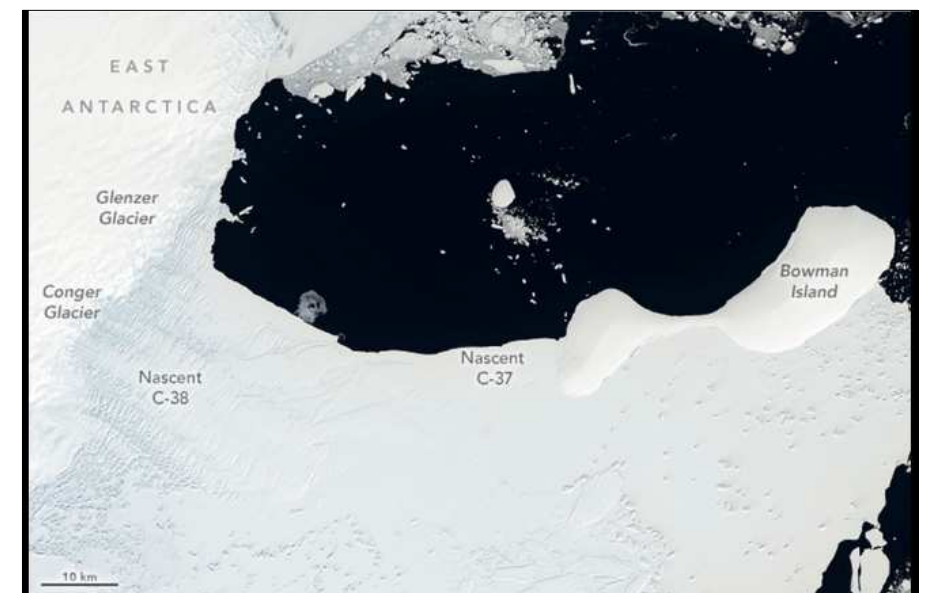


The Ice of Today

- Wave activity increasing in the polar regions
- Sea ice attenuates ocean wave propagation in both the Arctic and Antarctica
- Ocean waves have been an important factor in the recent ice shelf breakups, including the Conger-Glenzer
- Interesting physical phenomena generated by multiscale structure; geometric singularities; nonlocal effects; and infinite interfaces.

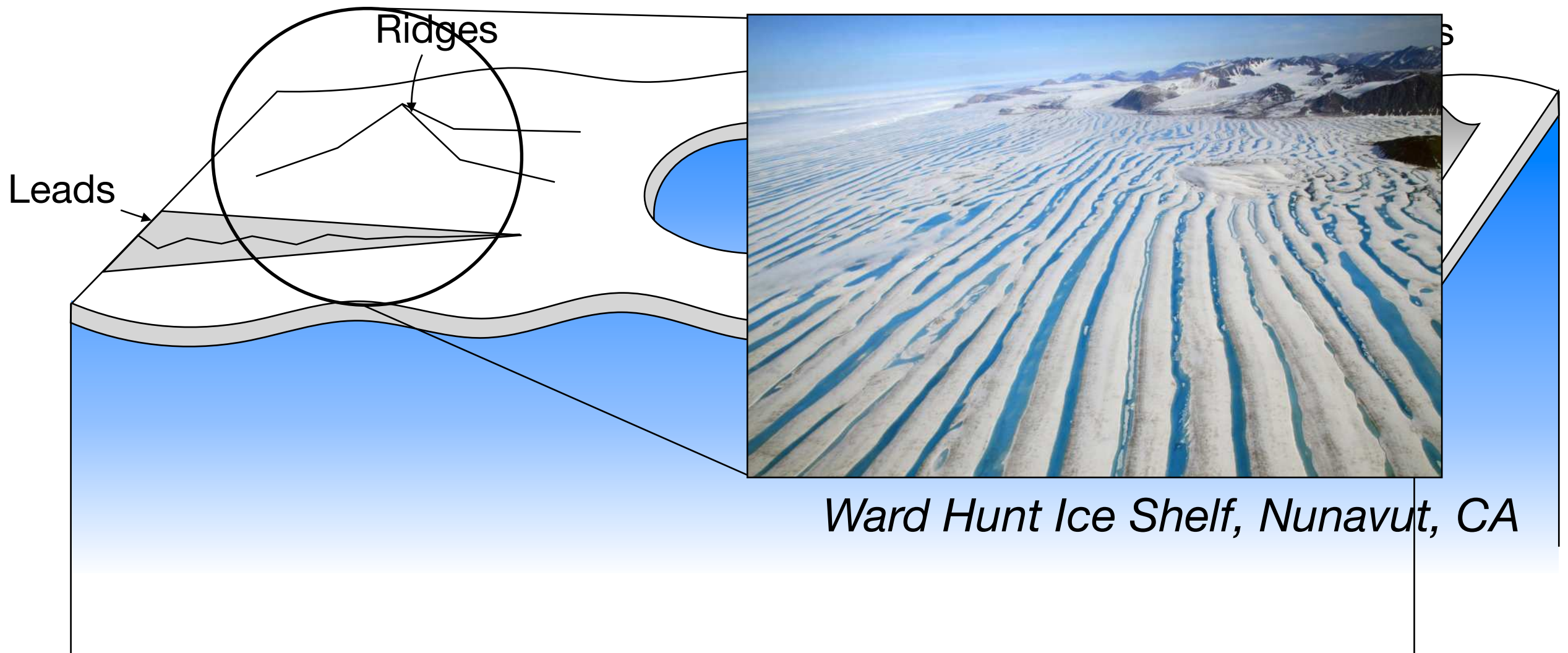


Average sea ice occurrence in the polar regions.

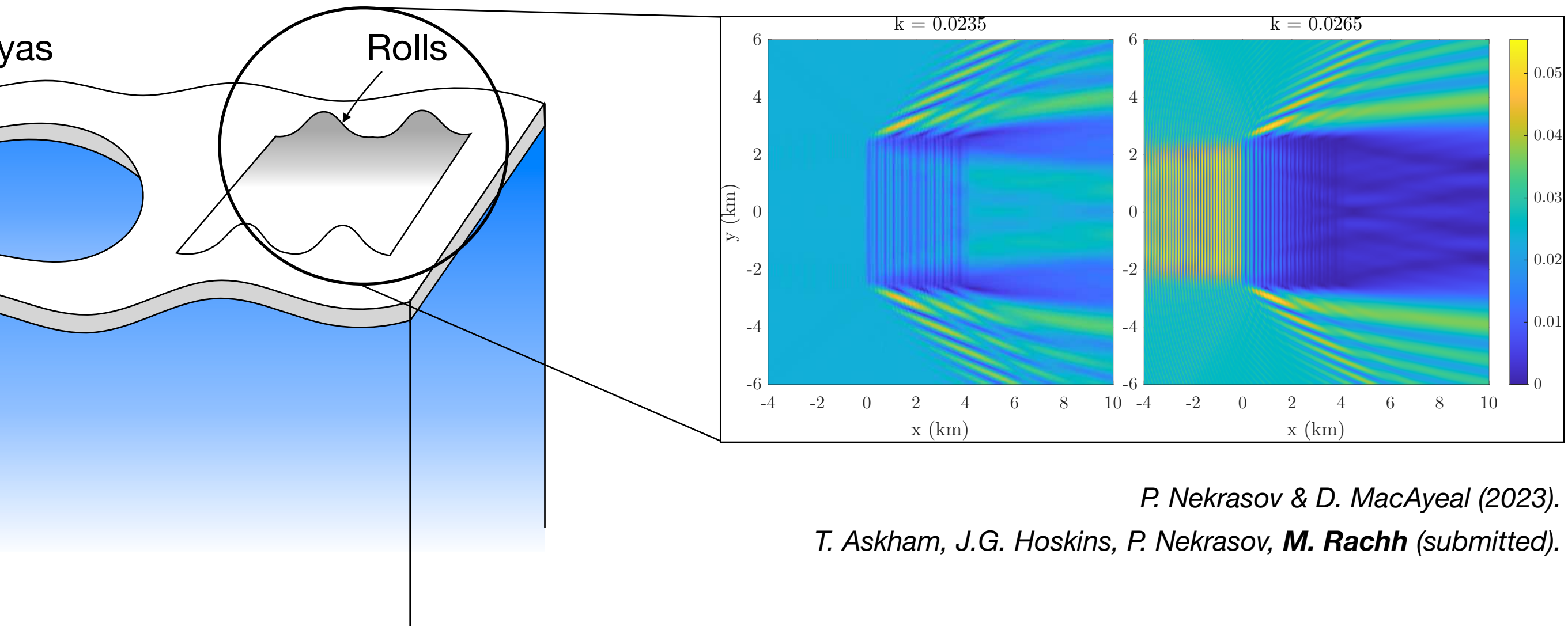


Collapse of Conger-Glenzer ice shelf (2022).

The complex morphology of ice



The complex morphology of ice



P. Nekrasov & D. MacAyeal (2023).

*T. Askham, J.G. Hoskins, P. Nekrasov, **M. Rachh** (submitted).*

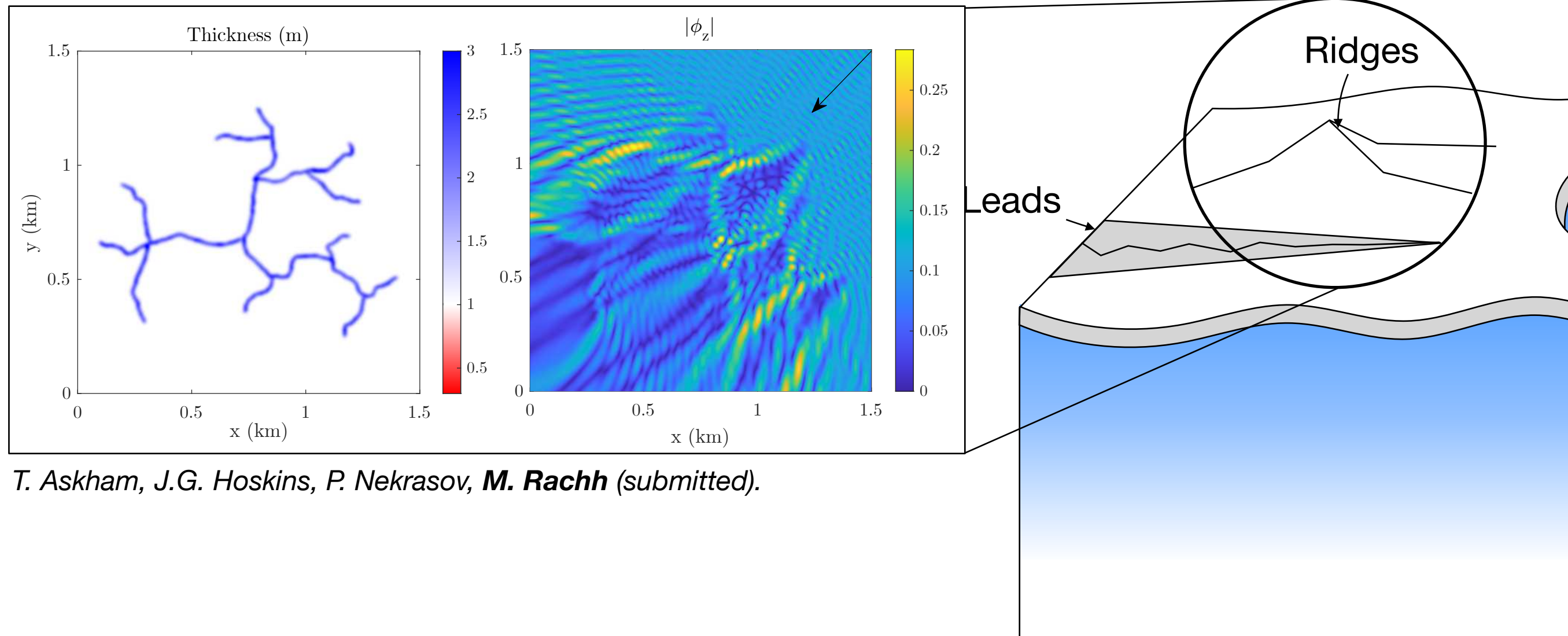
The complex morphology of ice

gas



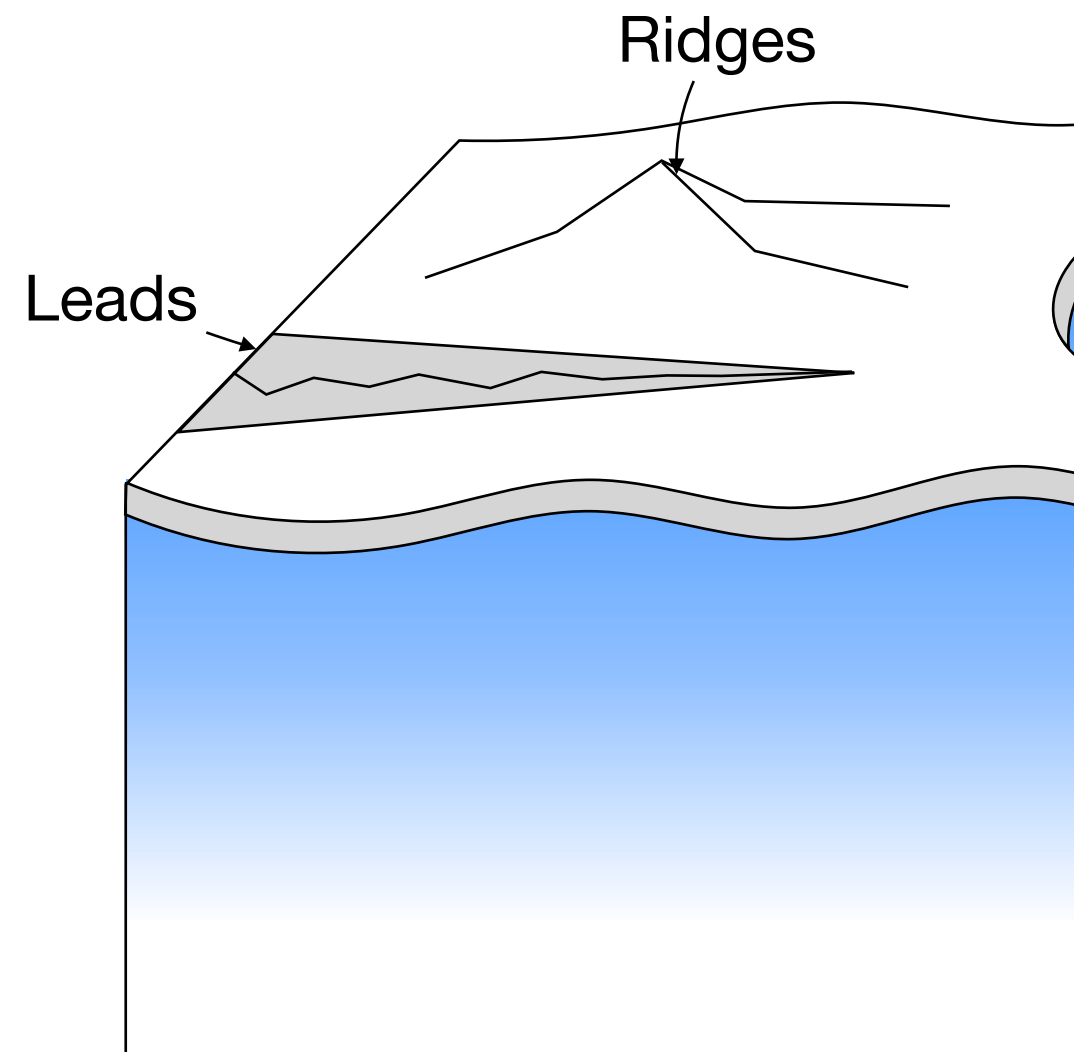
White Island, ANT

The complex morphology of ice

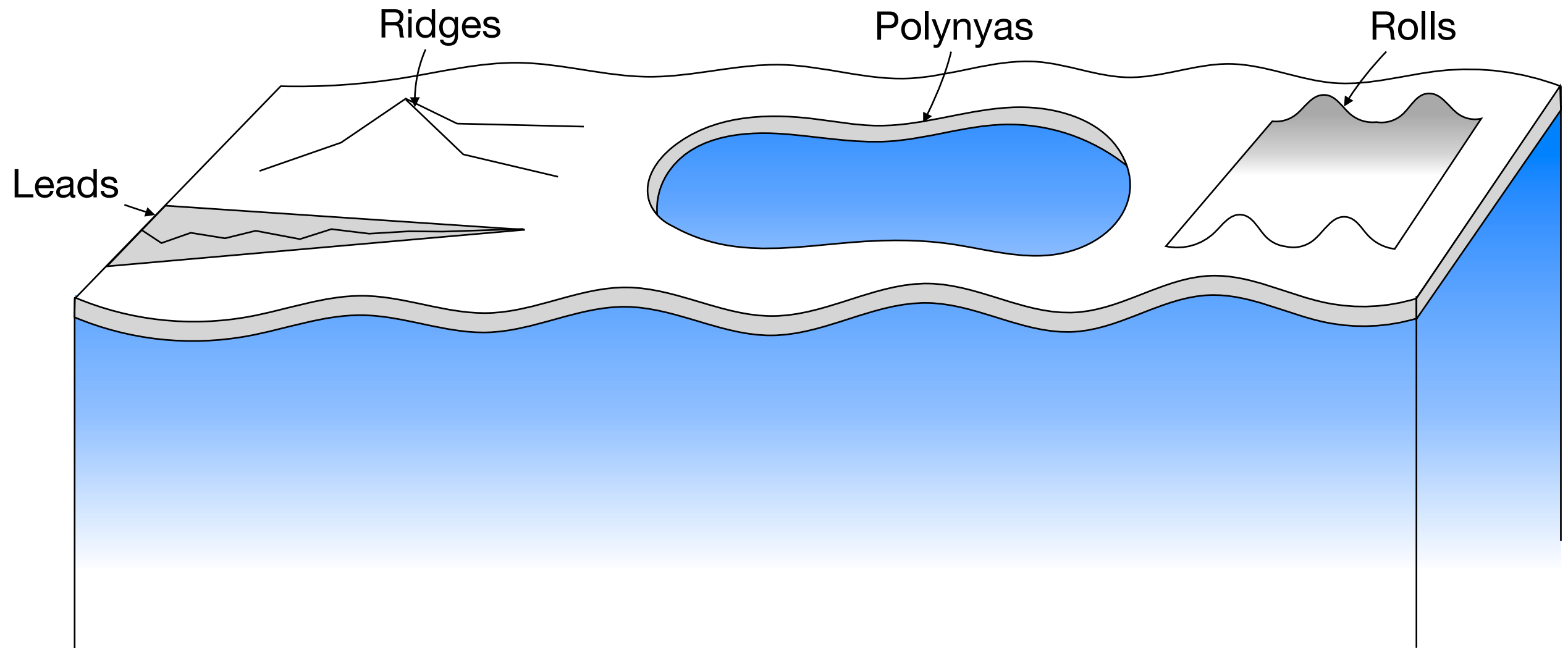


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The complex morphology of ice



The complex morphology of ice



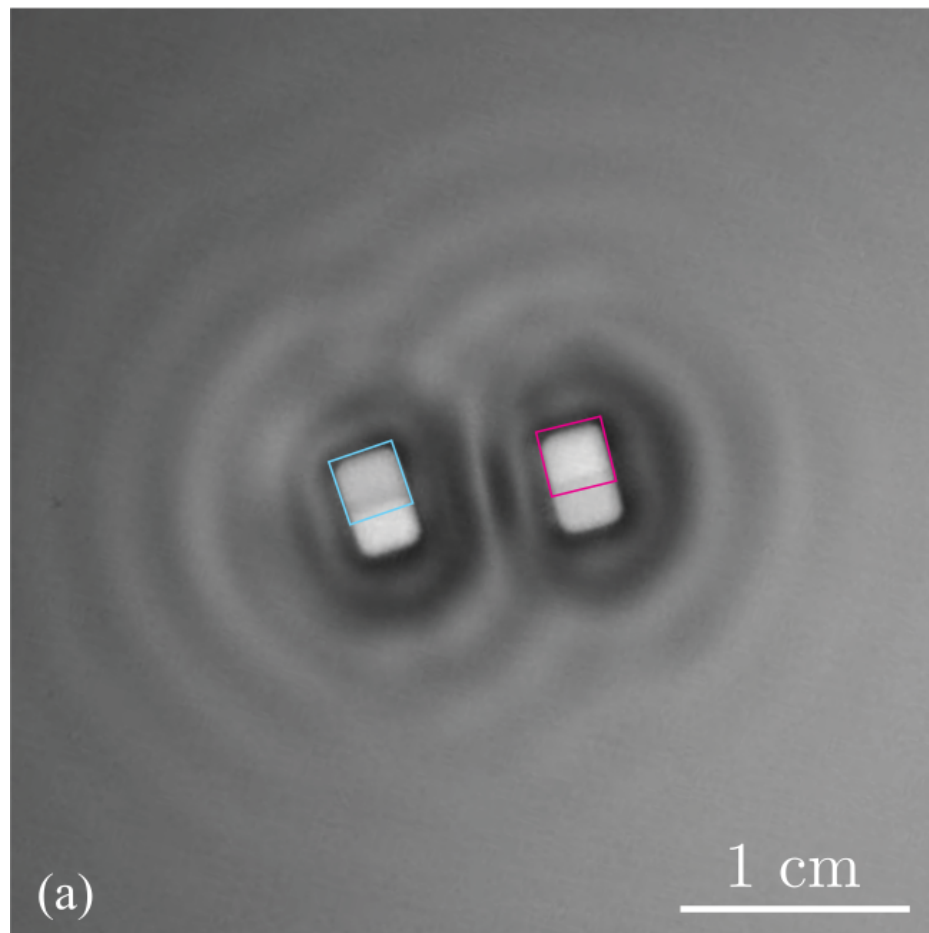
Linear gravity wave models



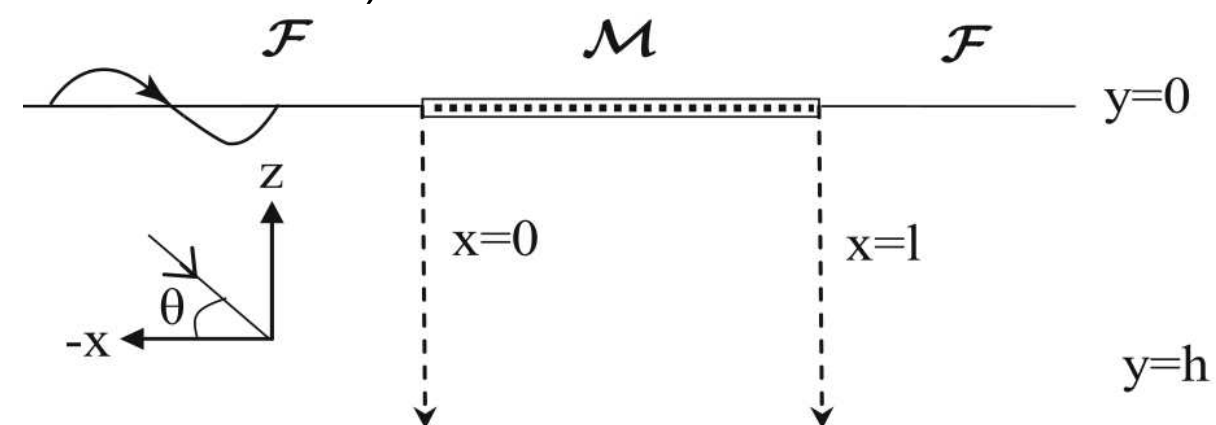
“... receive the first part of this paragraph as an apology on the part of mathematicians for applying to the motion of Tides a theory so self-evidently inadequate as the Equilibrium-theory.”

- *Tides and Waves* 1841
- Incompressible, irrotational, inviscid fluid → potential flow
- Small amplitude waves → pose problems in fixed domains
- Gravity waves are affected by changes in surface properties. Reynolds (1880) observed that an oil slick attenuates waves
- Augmented models with flexural effects (Greenhill 1886), surface tension (Thomson 1871), membrane, etc.

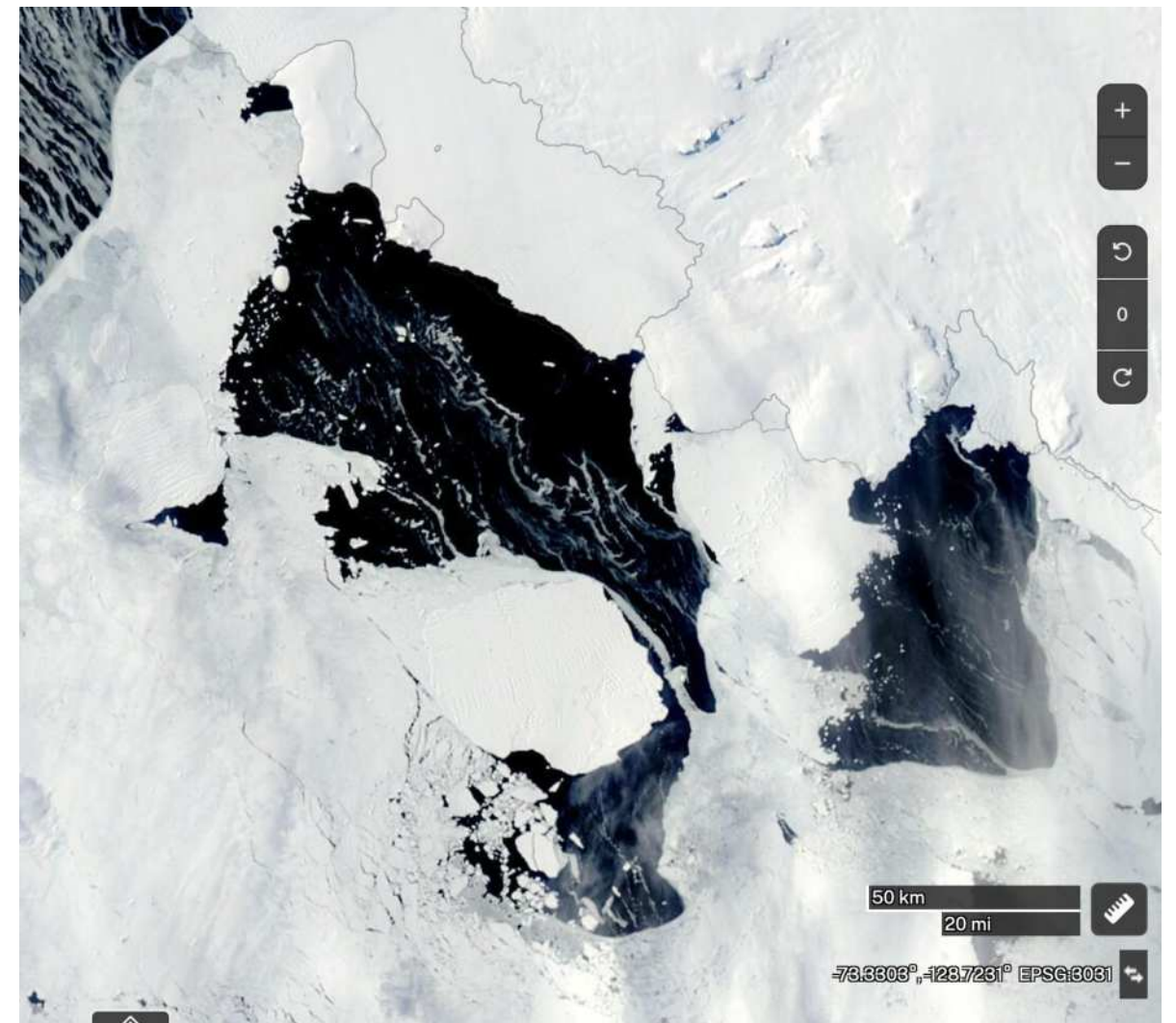
Applications across scales



~ 1cm: Capillary surfer (Oza, Pucci, Ho, and Harris 2017)



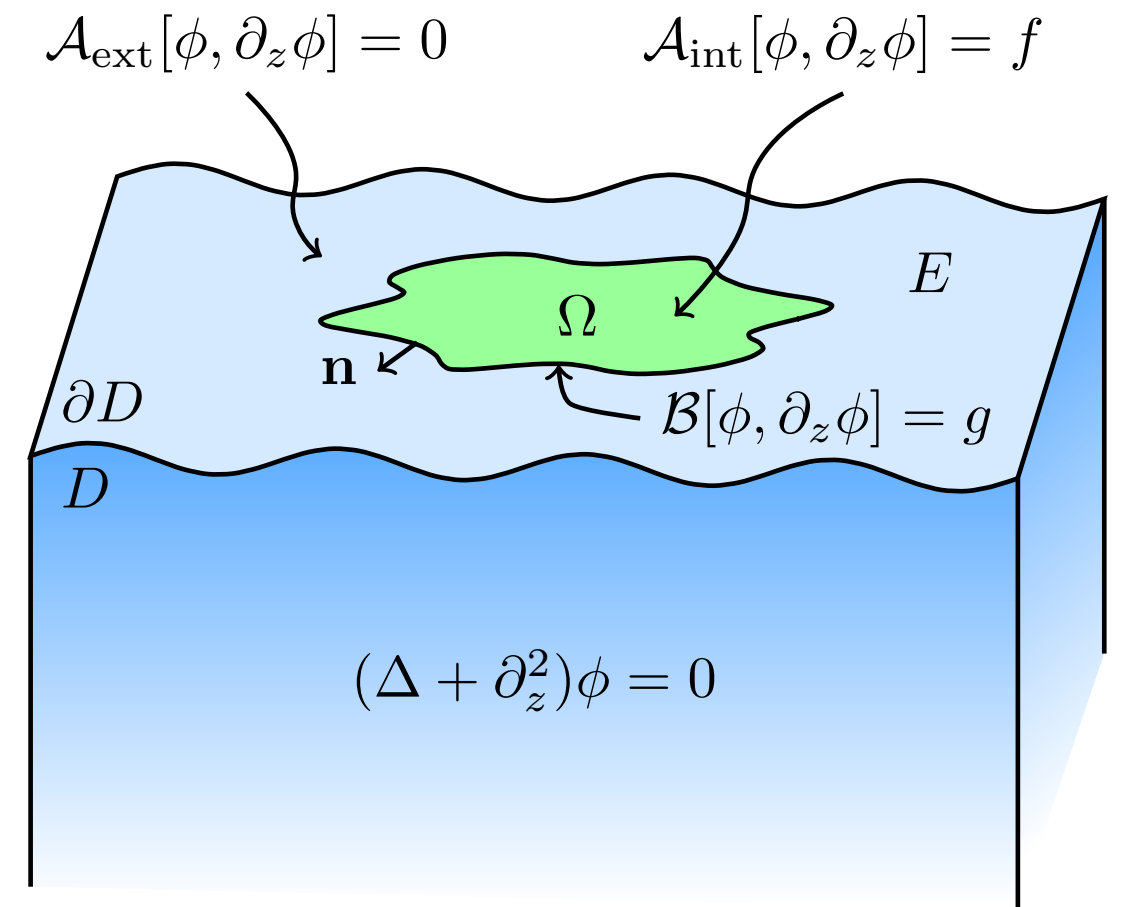
~ 30m: Porous membrane (Koley and Sahoo 2017)



~ 200km: Amundsen Sea Polynya (NASA Terra, Oct 25, 2024)

Bulk-surface problems for linear gravity waves

- Simple potential flow model in the bulk D , $\vec{u} = \nabla_{3D}\phi$
- $\mathcal{A}$ operators are boundary traces with surface differential operators, accounting for gravitational/tensile/elastic/flexural effects
- $\mathcal{B}$ operators are in-surface interface conditions



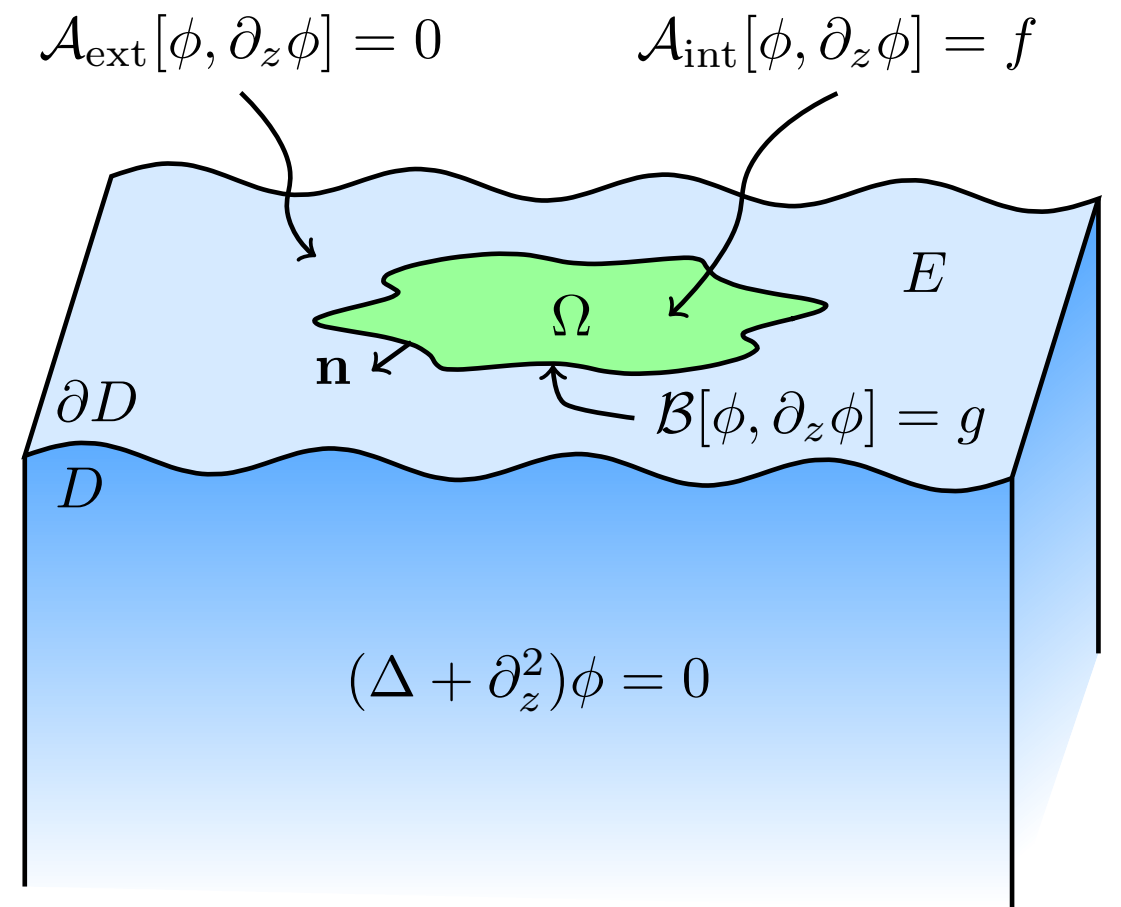
D : half-space, Ω : compact

$$\mathcal{A}[\phi, \partial_z \phi] = \underbrace{(a_2 \Delta^2)}_{\text{flexing}} - \underbrace{a_1 \Delta}_{\text{tension}} + \underbrace{a_0}_{\text{Inertia/gravity}} \partial_z \phi - \underbrace{\phi}_{\text{pressure}}$$

displacement

Considerations

$$\mathcal{A}[\phi, \partial_z \phi] = (a_2 \Delta^2 - a_1 \Delta + a_0) \partial_z \phi - \phi$$

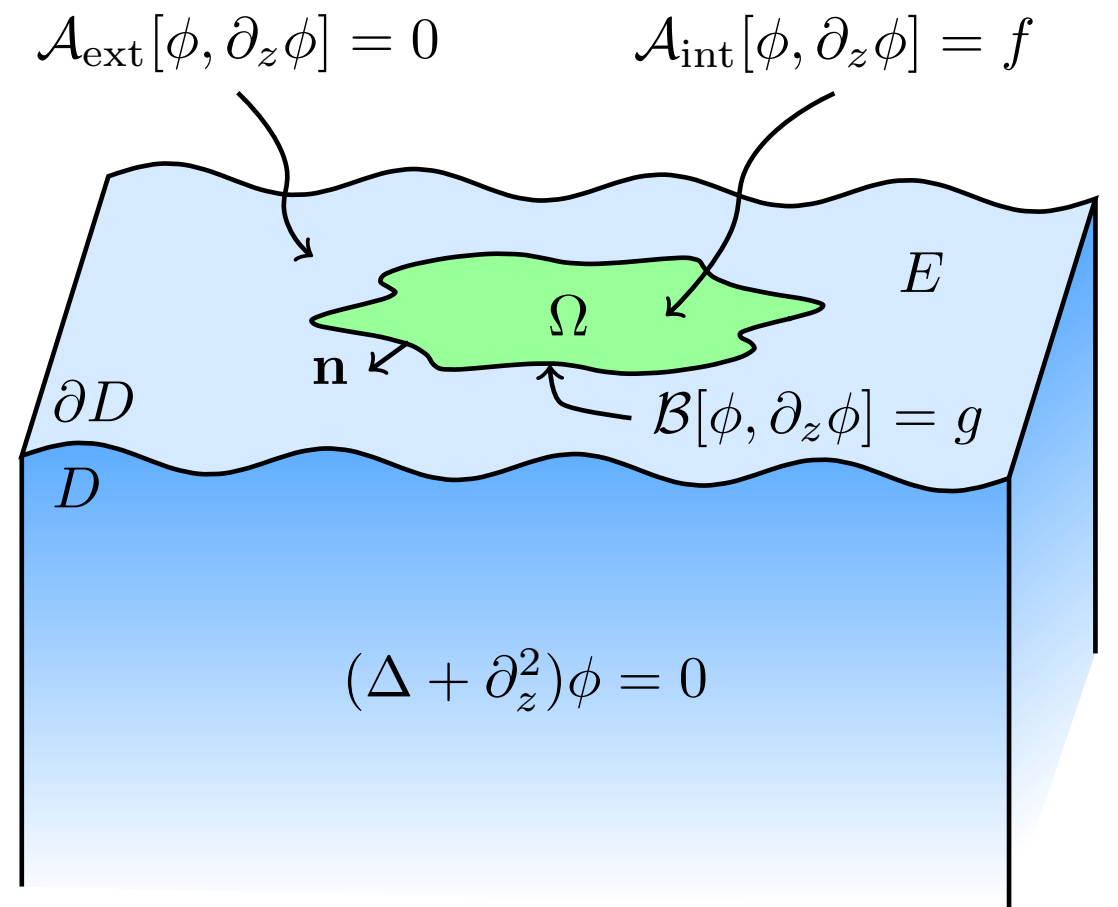


- Unbounded domain and (up-to) fourth order boundary operator $\rightarrow$ problems for PDE discretization
- Unbounded boundary $\rightarrow$ problems for boundary integral discretization
- Everything is linear, equation is elliptic
- *Morally*, should be able to reduce to a problem on the support of the data, i.e. Ω

Bulk $\rightarrow$ surface

$$\phi(r_{3d}) = p \cdot v \cdot \int_D \frac{1}{4\pi |r_{3d} - r'_{3d}|} \sigma(r') dA(r')$$

$$\mathcal{S}[\sigma](r_{3d})$$



- ϕ automatically harmonic in $z < 0$ for any σ
- Recall jump condition:

$$\begin{aligned} \lim_{z \rightarrow 0} \partial_z \mathcal{S}_{3D}[\sigma](r_{3d}) &= \frac{1}{2} \sigma(r) + p \cdot v \cdot \int_{\mathbb{R}^2} \frac{-z}{4\pi |r_{3d} - r'_{3d}|^3} \bigg|_{z=0} \sigma(r') dA(r') \\ &= \frac{1}{2} \sigma(r) \end{aligned}$$

Bulk $\rightarrow$ surface

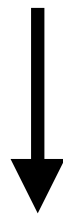
Ansatz: $\phi = \mathcal{S}_{3D}[\sigma] \quad \partial_z \phi = \sigma/2$

Boundary conditions:

$$(a_0 \Delta^2 - a_1 \Delta + a_0) \partial_z \phi - \phi = 0 \quad \text{in } E$$

$$\mathcal{A}_{int}[\phi, \partial_z \phi] = f \quad \text{in } \Omega$$

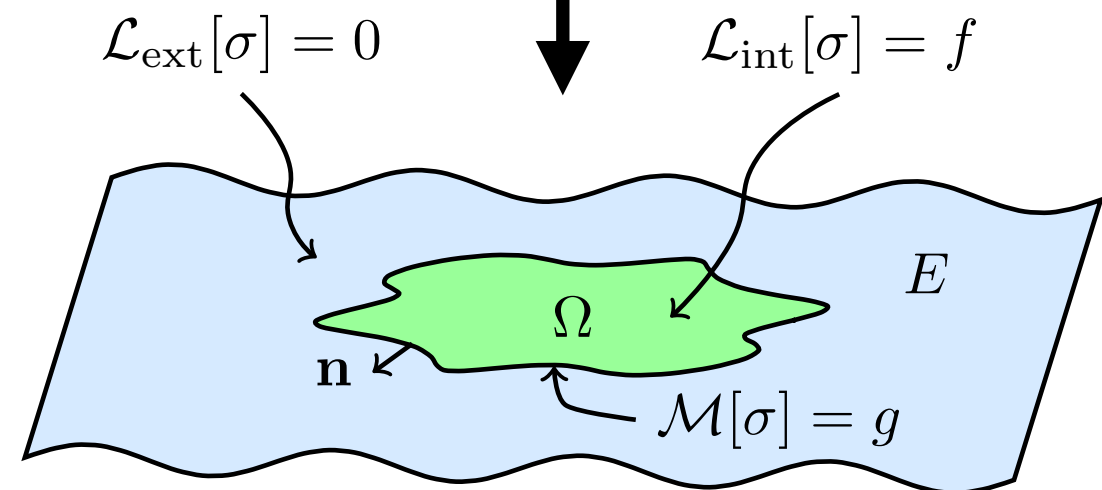
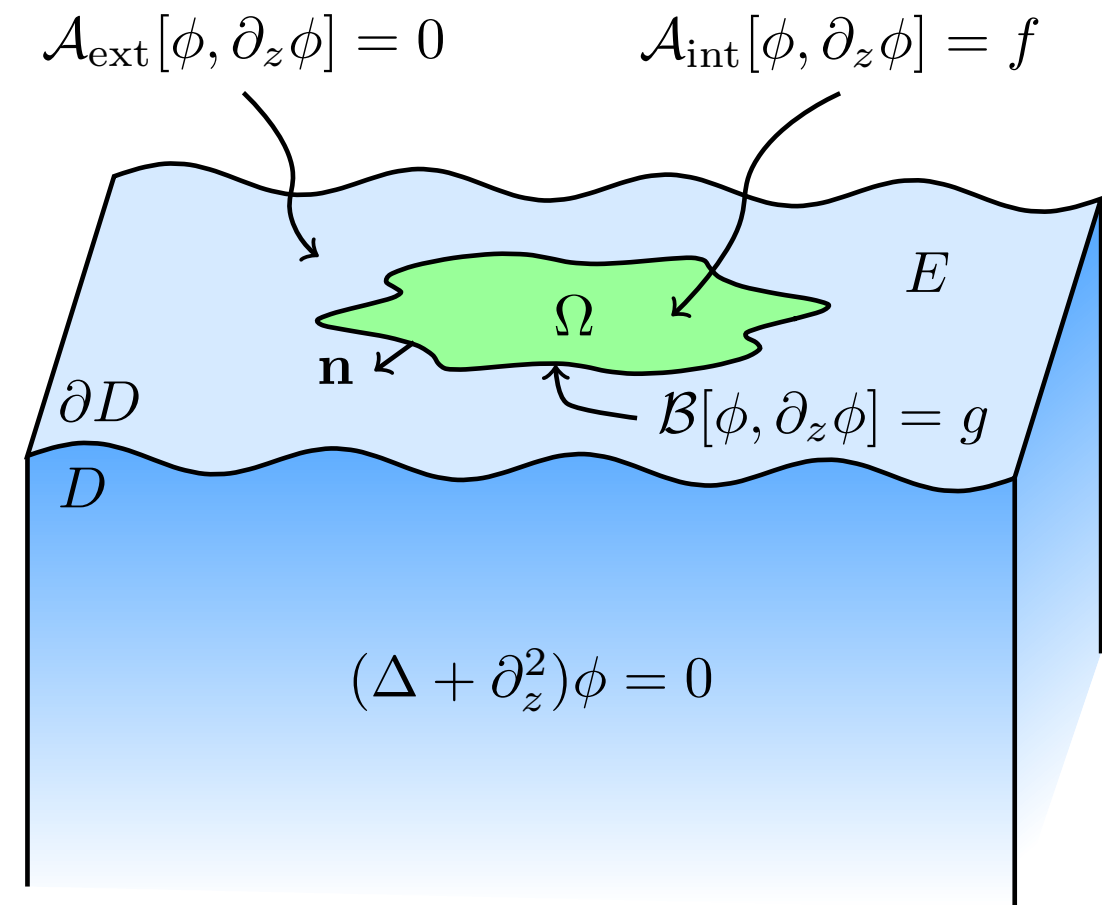
$$\mathcal{B}[\phi, \partial_z \phi] = g \quad \text{on } \partial\Omega$$



$$(a_0 \Delta^2 - a_1 \Delta + a_0) \sigma/2 - \mathcal{S}_{3D}[\sigma] = 0 \quad \text{in } E$$

$$\mathcal{L}_{int}[\sigma] := \mathcal{A}_{int}[\mathcal{S}_{3D}[\sigma], \sigma/2] = f \quad \text{in } \Omega$$

$$\mathcal{M}[\sigma] := \mathcal{B}[\mathcal{S}_{3D}[\sigma], \sigma/2] = g \quad \text{on } \partial\Omega$$



Bulk $\rightarrow$ surface

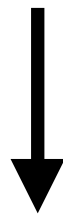
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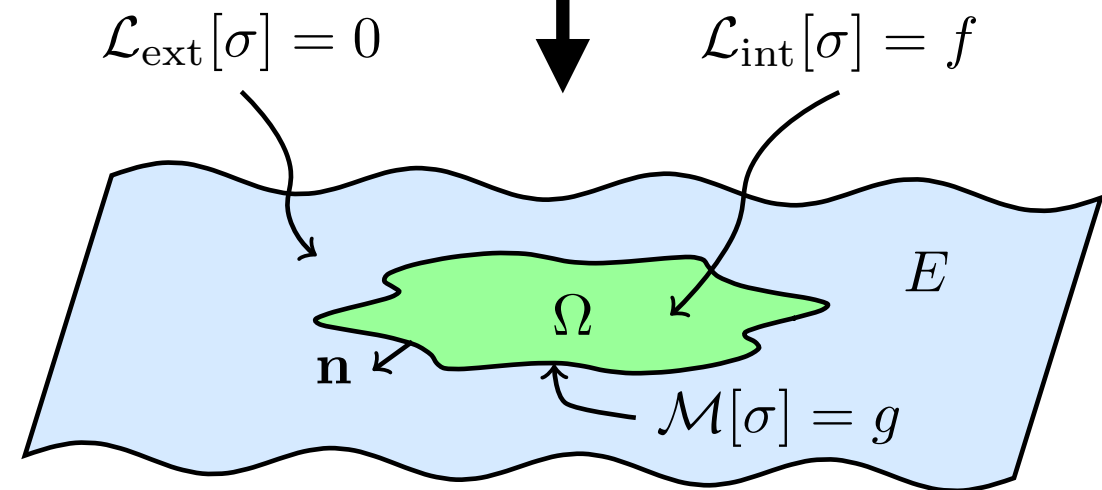
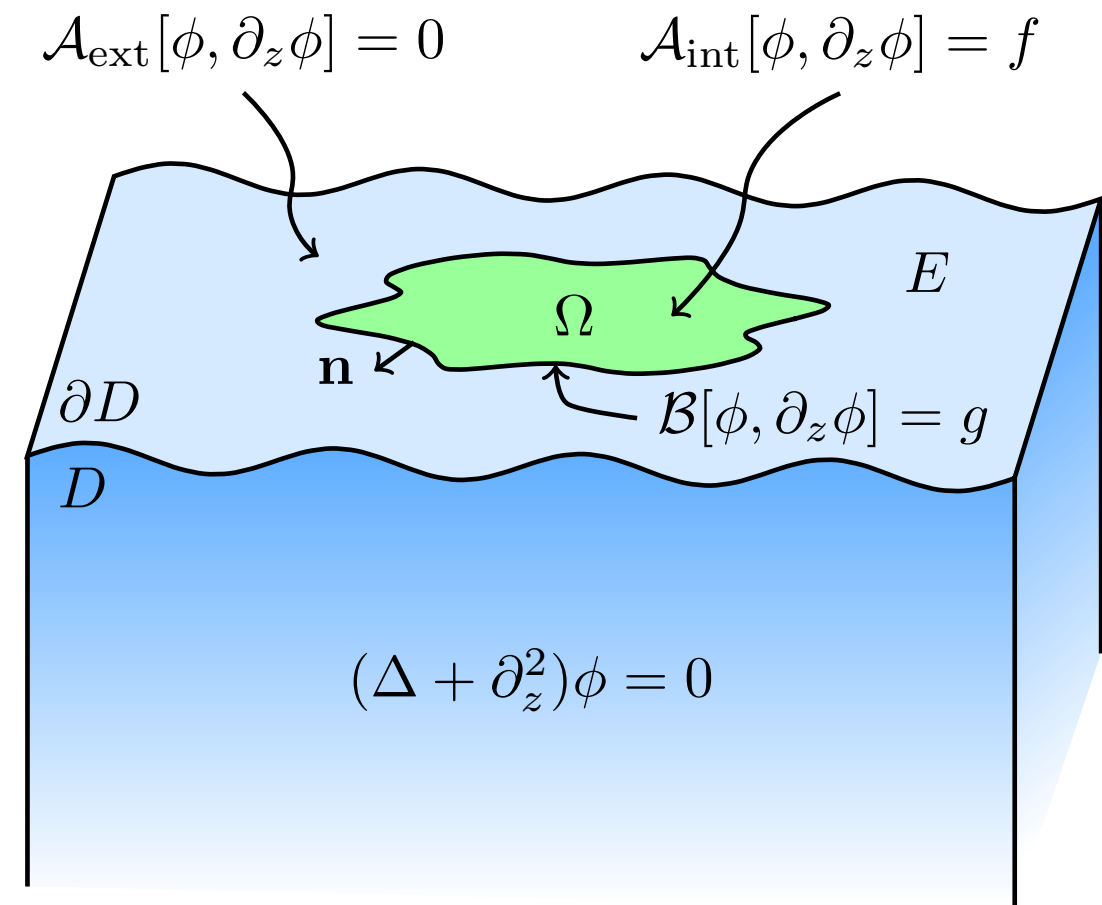


$$(a_0 \Delta^2 - a_1 \Delta + a_0) \sigma/2 - \mathcal{S}_{3D}[\sigma] = 0 \quad \text{in } E$$

$$\mathcal{L}_{int}[\sigma] := \mathcal{A}_{int}[\mathcal{S}_{3D}[\sigma], \sigma/2] = f \quad \text{in } \Omega$$

$$\mathcal{M}[\sigma] := \mathcal{B}[\mathcal{S}_{3D}[\sigma], \sigma/2] = g \quad \text{on } \partial\Omega$$

Challenge: Non-local term implies σ non-compactly supported and decays slowly at ∞



General framework

$$\mathcal{L}_{ext}[\sigma] := (a_0\Delta^2 - a_1\Delta + a_0)\sigma/2 - \mathcal{S}_{3D}[\sigma] = 0 \quad \text{in } E$$

$$\mathcal{L}_{int}[\sigma] := \mathcal{A}_{int}[\mathcal{S}_{3D}[\sigma], \sigma/2] = f \quad \text{in } \Omega$$

$$\mathcal{M}[\sigma] := \mathcal{B}[\mathcal{S}_{3D}[\sigma], \sigma/2] = g \quad \text{on } \partial\Omega$$

Reduce to an integral equation on $\Omega, \partial\Omega$ by making an ansatz:

$$\sigma(r) = \underbrace{\int_{\Omega} V(r, r')\mu(r')dA(r')}_{\text{surface-volume potential}} + \sum_{j=1}^m \underbrace{\int_{\partial\Omega} B_j(r, r')\rho_j(r')ds(r')}_{\text{surface-layer potential}}$$

V and B_j linear combinations of G_S and its derivatives:

$$\mathcal{L}_{ext}[G_S(\cdot, r')](r) = \delta(r - r')$$

- Equation in E automatically satisfied
- Design V and B_j so that applying $\mathcal{L}_{int}$ and $\mathcal{M}$ results in Fredholm equation of second kind (identity + compact)

G_S

$$p(-\Delta)G_S(r - r') - p \cdot v \cdot \int_{\mathbb{R}^2} \frac{1}{2\pi |r - r''|} G_S(r' - r'') dA(r'') = 2\delta(r - r')$$

$$p(z) = a_2 z^2 + a_1 z + a_0$$

Fourier
transform



$$\begin{aligned} \widehat{G}_S(\xi) &= \frac{2|\xi|}{|\xi|p(|\xi|^2) - 1} \\ &= 2|\xi| \sum_{j=1}^{deg(q)} \frac{e_j}{|\xi| - \rho_j} \\ &= 2 \sum_{j=1}^{deg(q)} \frac{e_j \rho_j}{|\xi| - \rho_j} \end{aligned}$$

$$q(z) = zp(z^2) - 1$$

ρ_j roots of q , and e_j residue of $1/q$ at ρ_j

$$\sum_{j=1}^{deg(q)} e_j \rho_j^\ell = 0 \text{ for } \ell = 0, 1 \text{ if } deg(q) \geq 3$$

G_S

$$\widehat{G}_S(\xi) = 2 \sum_{j=1}^{\deg(q)} \frac{e_j \rho_j}{|\xi| - \rho_j}$$

Identity for $\rho_j \notin \mathbb{R}_+$

$$\mathcal{F}^{-1} \left[\frac{1}{|\xi - \rho_j|} \right] = \frac{1}{2\pi |r|} + \frac{\rho_j}{4} \mathbf{K}_0(-\rho_j |r|)$$

$a_2, a_1 \geq 0$ and $\deg(q) \geq 3$, $\rho_1 \in \mathbb{R}_+$. Limiting absorption gives
us

$$G_S(r, r') = \frac{1}{2} \left(e_1 \rho_1^2 \left[-\mathbf{K}_0(\rho_1 |r - r'|) + 2iH_0^{(1)}(\rho |r - r'|) \right] + \sum_{j=2}^{\deg(q)} e_j \rho_j^2 \mathbf{K}_0(-\rho_j |r - r'|) \right)$$
$$\mathcal{S}_{3D}[G_S] = \frac{1}{2} \left(e_1 \rho_1 \left[-\mathbf{K}_0(\rho_1 |r - r'|) + 2iH_0^{(1)}(\rho |r - r'|) \right] + \sum_{j=2}^{\deg(q)} e_j \rho_j \mathbf{K}_0(-\rho_j |r - r'|) \right)$$

G_S

$$G_S(r, r') = \frac{1}{2} \left(e_1 \rho_1^2 \left[-\mathbf{K}_0(\rho_1 |r - r'|) + 2iH_0^{(1)}(\rho |r - r'|) \right] + \sum_{j=2}^{deg(q)} e_j \rho_j^2 \mathbf{K}_0(-\rho_j |r - r'|) \right)$$

- $H_0^{(1)}$ radiates like Helmholtz equation, i.e.

$$\frac{r}{|r|} \cdot \nabla G_S - i\rho_1 G_S = o\left(\frac{1}{\sqrt{|r|}}\right)$$

- Remaining terms have $1/|r|^3$ decay

- $|r - r'| \rightarrow 0, \quad G_S(r, r') \sim \begin{cases} |r - r'|^2 \log |r - r'| & deg(p) = 2 \\ \log |r - r'| & deg(p) = 1 \end{cases}$

Partial membranes/Capillary surfers

Ansatz: $\phi = \mathcal{S}_{3D}[\sigma] \quad \partial_z \phi = \sigma/2$

Boundary conditions:

$$\mathcal{A}_{ext} := (-\beta\Delta + \gamma)\partial_z \phi - \phi = 0 \quad \text{in } E$$

$$\mathcal{A}_{int} := \partial_z \phi - \phi = f \quad \text{in } \Omega$$

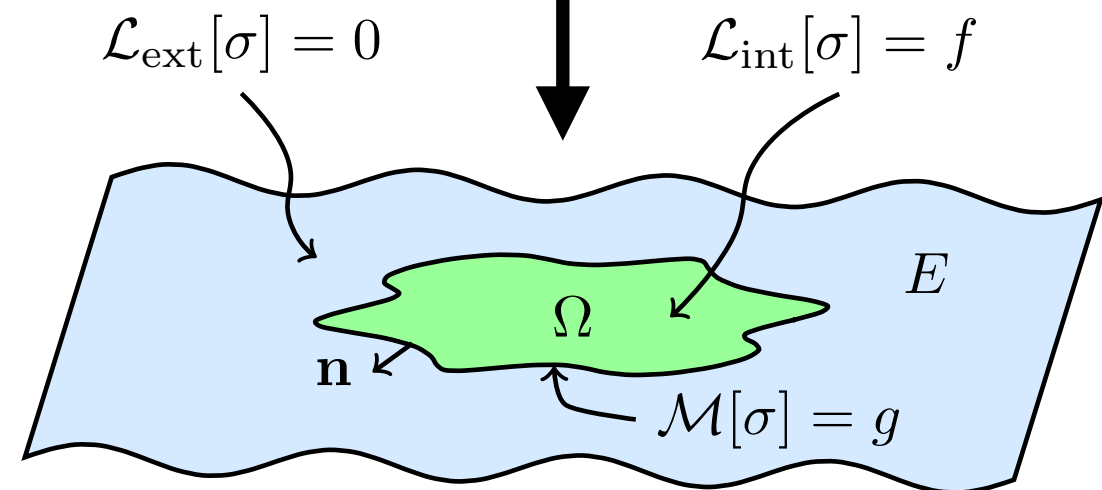
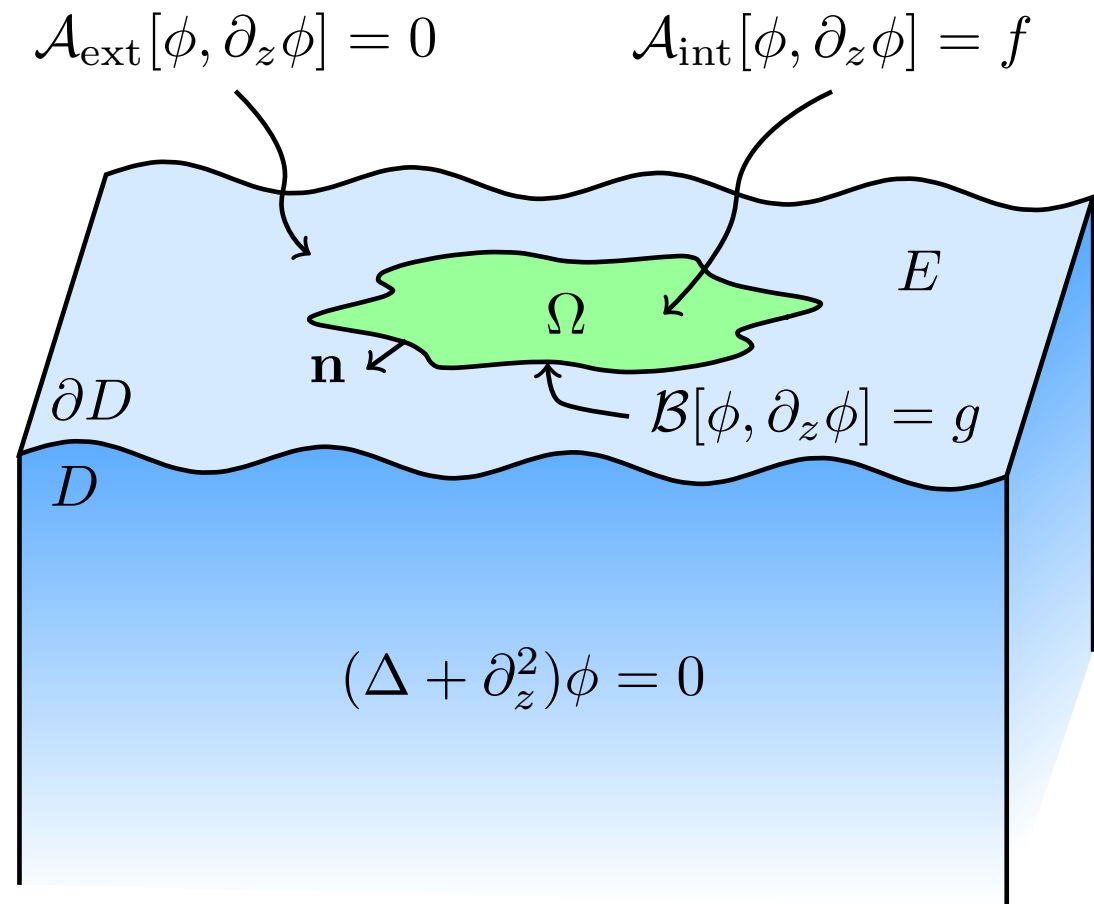
$$\mathcal{B} := \gamma_0^+ \partial_n \partial_z \phi = g \quad \text{on } \partial\Omega$$



$$\mathcal{L}_{ext} := (-\beta\Delta + \gamma)\sigma/2 - \mathcal{S}_{3D}[\sigma] = 0 \quad \text{in } E$$

$$\mathcal{L}_{int} := \sigma/2 - \mathcal{S}_{3D}[\sigma] = f \quad \text{in } \Omega$$

$$\mathcal{M} := \gamma_0^+ \partial_n \sigma/2 = g \quad \text{on } \partial\Omega$$



Integral representation

$$\sigma(r) = \int_{\Omega} V(r, r') \mu(r') dA(r') + \int_{\partial\Omega} B(r, r') \tau_j(r') ds(r')$$

How to choose V and B ?

Recall: $\frac{1}{2} (-\beta\Delta + \gamma) G_S(r, r') - \mathcal{S}_{3D}[G_S(\cdot, r')](r) = \delta(r - r')$

G_S a smooth-ish perturbation of 2-D Laplace Green's function

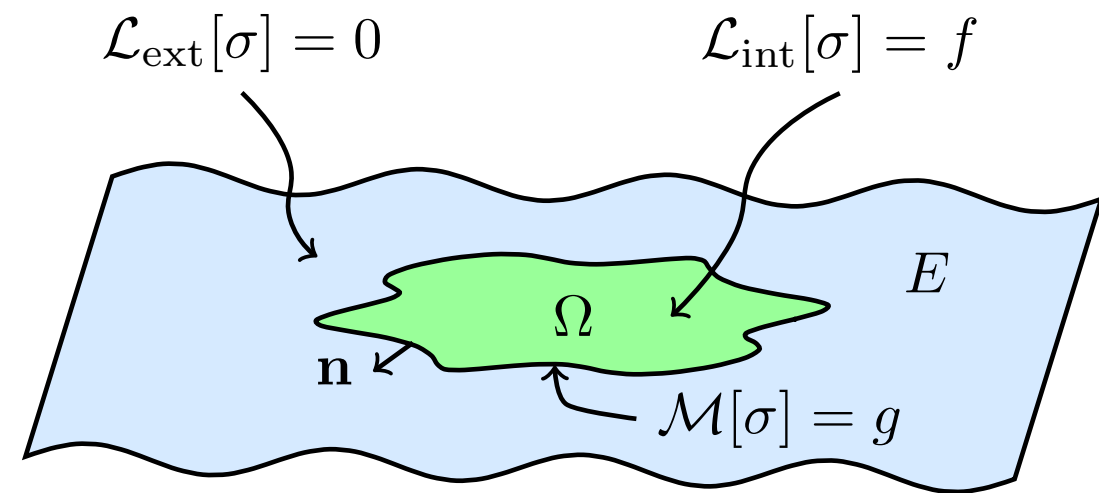
$$V(r, r') = -\frac{\beta}{2} \Delta_{r'} G_S(r, r') + \frac{|\gamma|}{2} G_S(r, r') \quad \text{Identity term in volume eqn} \checkmark$$

$$= \delta(r - r') + \frac{|\gamma| - \gamma}{2} G_S(r, r') + G_{\phi}(r, r')$$

$$B(r, r') = G_S(r, r') \quad \text{Identity term in boundary eqn} \checkmark$$

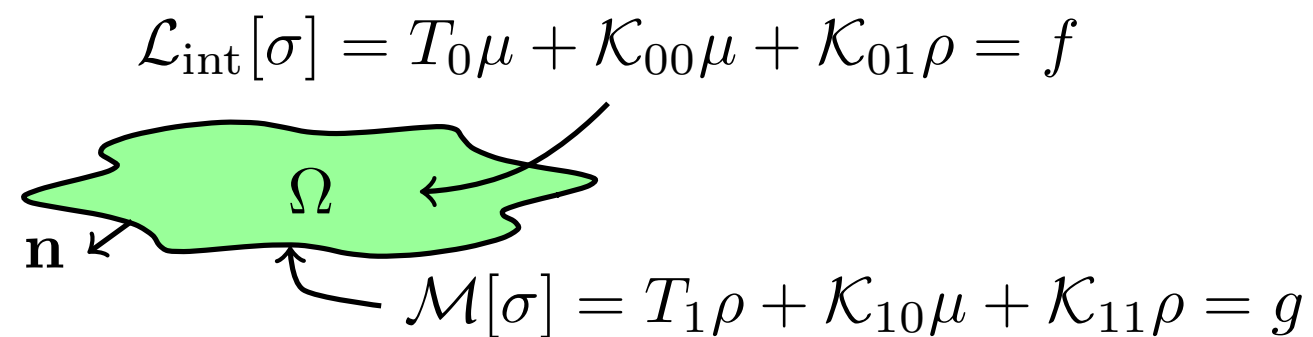
Similar to using single layer for Laplace Neumann problem

$$\begin{aligned}\mathcal{L}_{ext} &:= (-\beta\Delta + \gamma)\sigma/2 - \mathcal{S}_{3D}[\sigma] = 0 && \text{in } E \\ \mathcal{L}_{int} &:= \sigma/2 - \mathcal{S}_{3D}[\sigma] = f && \text{in } \Omega \\ \mathcal{M} &:= \gamma_0^+ \partial_n \sigma/2 = g && \text{on } \partial\Omega\end{aligned}$$



$$\sigma(r) = \int_{\Omega} V(r, r') \mu(r') dA(r') + \int_{\partial\Omega} B(r, r') \tau_j(r') ds(r')$$

Fredholm second kind system on
 $L^2(\Omega) \times L^2(\partial\Omega)^2$



$$\begin{aligned}K_{00} &= \frac{|\gamma| - \gamma}{4} G_S + \frac{1 - |\gamma|}{2} G_\phi + \frac{\beta}{2} \Delta_{r'} G_\phi & K_{01} &= \frac{1}{2} G_S - G_\phi & T_0 &= 1/2 \\ K_{10} &= \frac{|\gamma| - \gamma}{4} \partial_n G_S + \frac{1}{2} \partial_n G_\phi & K_{01} &= \frac{1}{2} \partial_n G_S & T_1 &= 1/(2\beta)\end{aligned}$$

K_{00}

$$\mathcal{L}_{int}[\sigma] = \frac{1}{2}\sigma - \mathcal{S}_{3D}[\sigma] = f$$

$$V(r, r') = \delta(r - r') + \frac{|\gamma| - \gamma}{2} G_S(r, r') + G_\phi(r, r')$$

$$\frac{1}{2}(-\beta\Delta + \gamma)G_S = G_\phi(r, r') + \delta(r - r')$$

$$\frac{1}{2}(-\beta\Delta + \gamma)G_\phi = \mathcal{S}_{3D}[G_\phi(r, r') + \delta(r - r')]$$

So

$$\mathcal{L}_{int}[V(r, r')] = \frac{1}{2}\delta(r - r') + \frac{|\gamma| - \gamma}{4} G_S(r, r') + \frac{1}{2} G_\phi(r, r')$$

$$- \mathcal{S}_{3D}[\delta(r - r')] - \frac{|\gamma| - \gamma}{4} G_\phi(r, r') - \mathcal{S}_{3D}[G_\phi]$$

$$= \frac{1}{2}\delta(r - r') + \frac{|\gamma| - \gamma}{4} G_S(r, r') + \frac{\beta}{2} \Delta_{r'} G_\phi(r, r') + \frac{1 - |\gamma|}{2} G_\phi(r, r')$$

Jump component

K_{00}

Polynya

Ansatz: $\phi = \mathcal{S}_{3D}[\sigma] \quad \partial_z \phi = \sigma/2$

Boundary conditions:

$$\mathcal{A}_{ext} := (\alpha \Delta^2 + \gamma) \partial_z \phi - \phi \quad \text{in } E$$

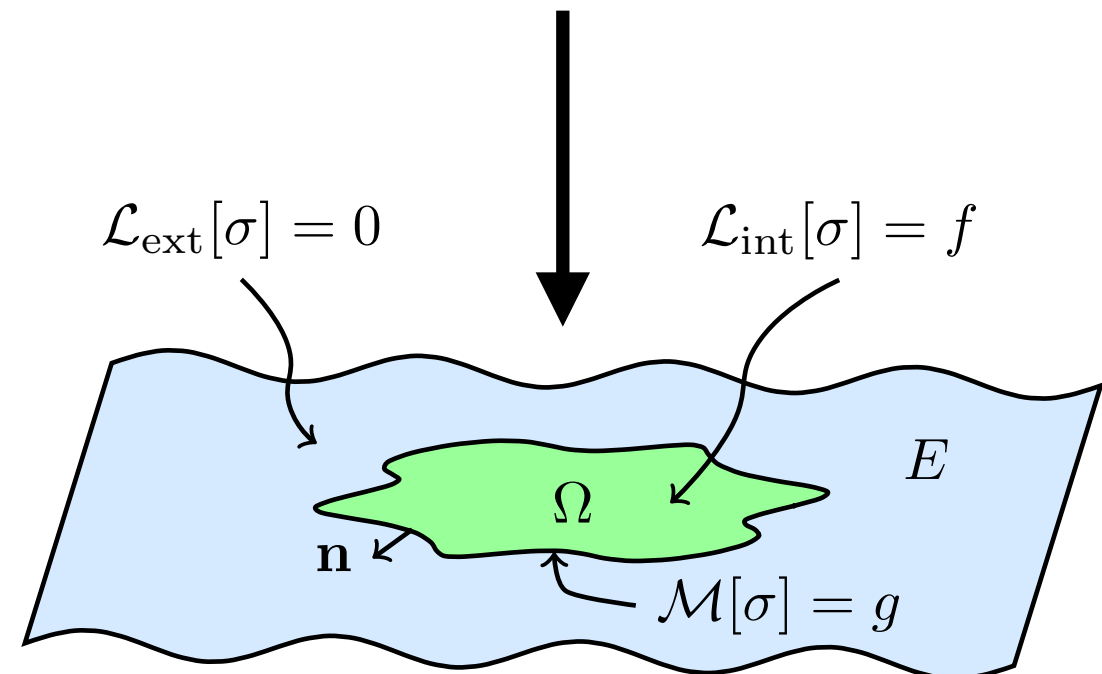
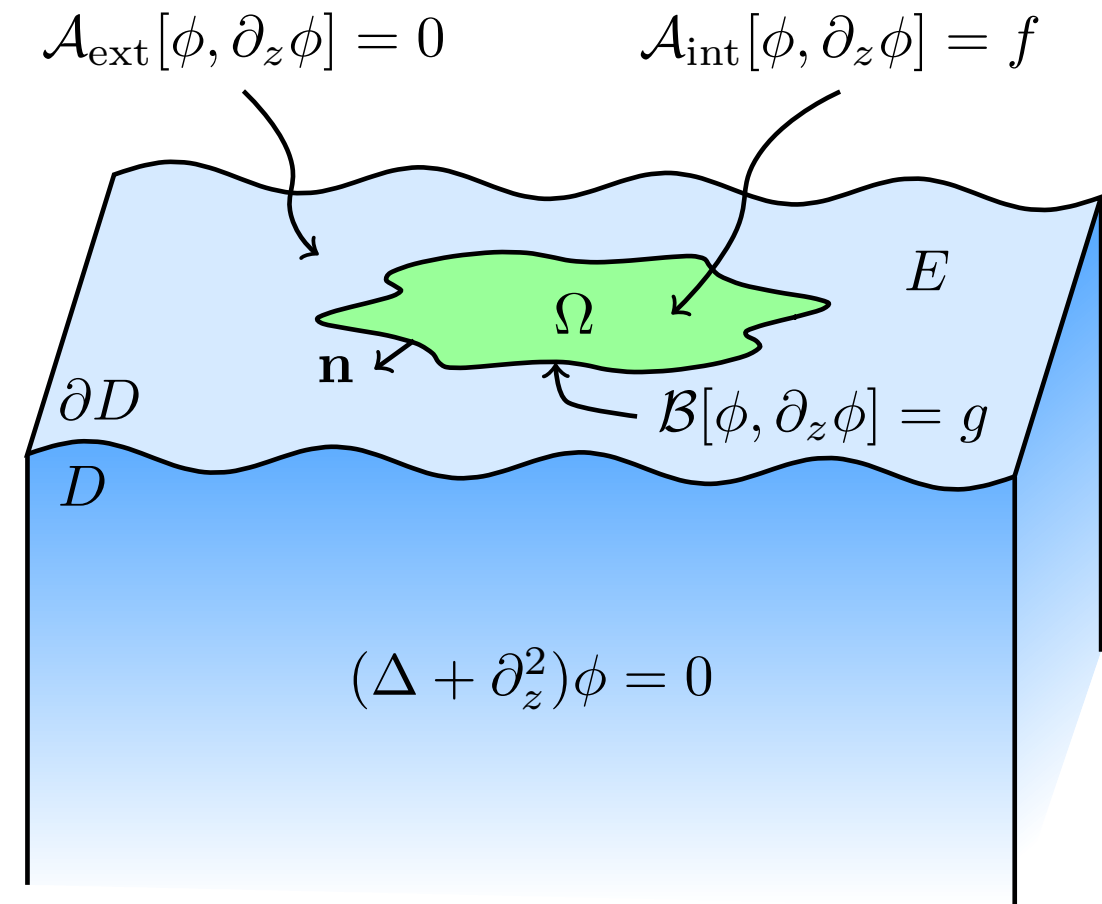
$$\mathcal{A}_{int} := \partial_z \phi - \phi \quad \text{in } \Omega$$

$$\mathcal{B} := \gamma_0^+ \begin{bmatrix} \nu \Delta \partial_z \phi + (1 - \nu) \partial_n^2 \partial_z \phi \\ \partial_n \Delta \partial_z \phi + (1 - \nu) \frac{d}{ds} \partial_{n\tau}^2 \partial_z \phi \end{bmatrix} \quad \text{on } \partial\Omega$$

$$\mathcal{L}_{ext} := (\alpha \Delta^2 + \gamma) \sigma/2 - \mathcal{S}_{3D}[\sigma] \quad \text{in } E$$

$$\mathcal{L}_{int} := \sigma/2 - \mathcal{S}_{3D}[\sigma] \quad \text{in } \Omega$$

$$\mathcal{M} := \gamma_0^+ \frac{1}{2} \begin{bmatrix} \nu \Delta \sigma + (1 - \nu) \partial_n^2 \sigma \\ \partial_n \Delta \sigma + (1 - \nu) \frac{d}{ds} \partial_{n\tau}^2 \sigma \end{bmatrix} \quad \text{on } \partial\Omega$$



Integral representation

$$\sigma(r) = \int_{\Omega} V(r, r') \mu(r') dA(r') + \int_{\partial\Omega} B(r, r') \tau_j(r') ds(r')$$

How to choose V and B ?

$$\text{Recall: } \frac{1}{2} (\alpha \Delta^2 + \gamma) G_S(r, r') - \mathcal{S}_{3D}[G_S(\cdot, r')](r) = \delta(r - r')$$

G_S a smooth-ish perturbation of 2-D Biharmonic Green's function

$$V(r, r') = \frac{\alpha}{2} \Delta_{r'}^2 G_S(r, r')$$

Identity term in volume eqn ✓

$$= \delta(r - r') - \frac{\gamma}{2} G_S(r, r') + G_{\phi}(r, r')$$

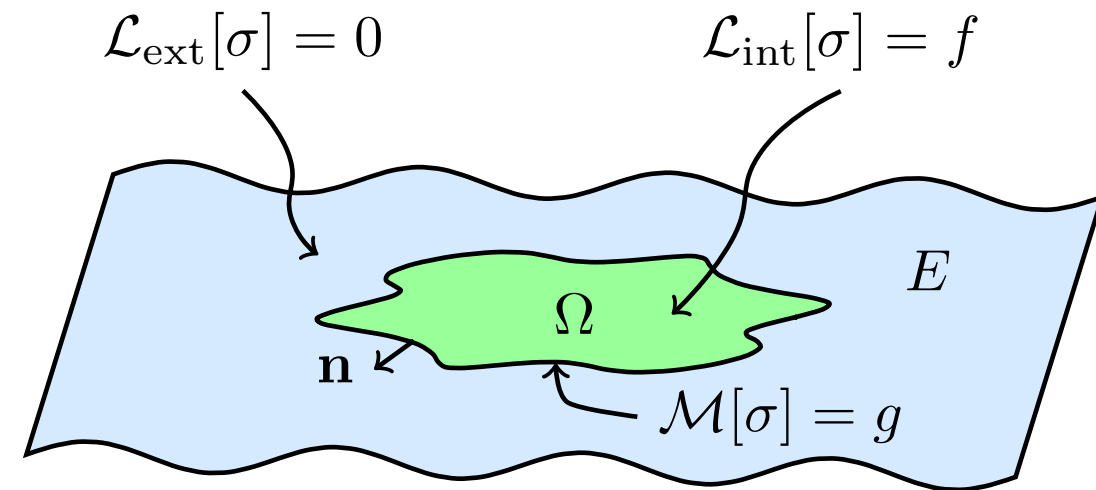
$$B(r, r') = \begin{bmatrix} \partial_n G_S + \eta \partial_{\tau'} G_S \star K_H \\ G_S \end{bmatrix}$$

Identity term in boundary eqn ✓

Similar to free plate flexural problem

Askham, Hoskins, Nekrasov, Su
arXiv 2409.19160

$$\begin{aligned}\mathcal{L}_{ext} &:= (\alpha\Delta^2 + \gamma)\sigma/2 - \mathcal{S}_{3D}[\sigma] && \text{in } E \\ \mathcal{L}_{int} &:= \sigma/2 - \mathcal{S}_{3D}[\sigma] && \text{in } \Omega \\ \mathcal{M} &:= \gamma_0^+ \frac{1}{2} \begin{bmatrix} \nu\Delta\sigma + (1-\nu)\partial_n^2\sigma \\ \partial_n\Delta\sigma + (1-\nu)\frac{d}{ds}\partial_{n\tau}^2\sigma \end{bmatrix} && \text{on } \partial\Omega\end{aligned}$$



$$\sigma(r) = \int_{\Omega} V(r, r')\mu(r')dA(r') + \int_{\partial\Omega} B(r, r')\tau_j(r')ds(r')$$

Fredholm second kind system on
 $L^2(\Omega) \times L^2(\partial\Omega)^2$

$$\begin{aligned}\mathcal{L}_{int}[\sigma] &= T_0\mu + \mathcal{K}_{00}\mu + \mathcal{K}_{01}\rho = f \\ \mathcal{M}_\ell[\sigma] &= T_\ell\rho_\ell + \mathcal{K}_{\ell 0}\mu + \sum_{j=1}^m \mathcal{K}_{\ell j}\rho_j = g\end{aligned}$$

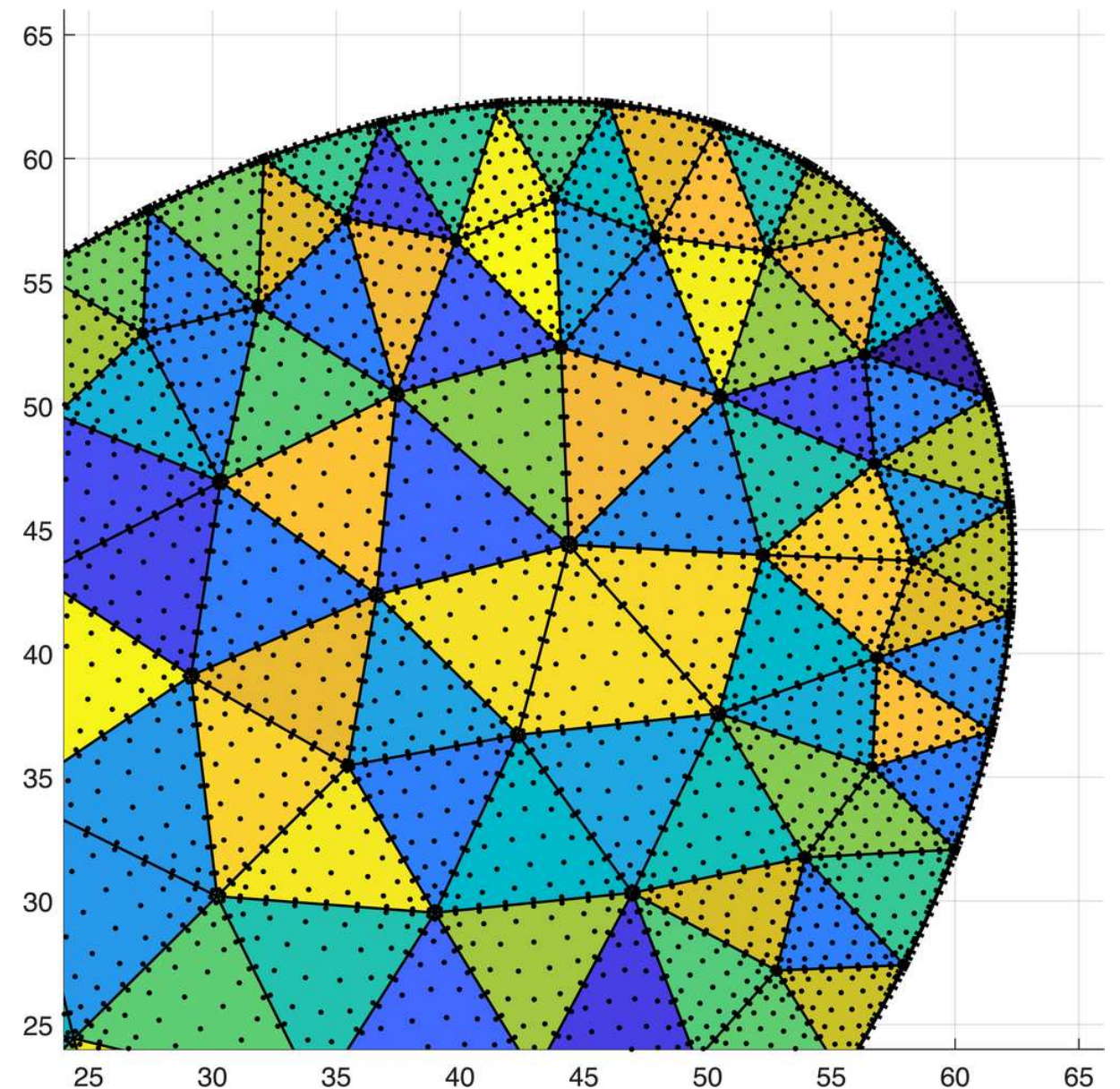
$$K_{00} = -\frac{\gamma}{4}G_S + \frac{1}{2}G_\phi - \frac{\alpha}{2}\Delta_{r'}^2 G_\phi$$

$$K_{10} = \frac{1}{2}\partial_{n'}G_S - \partial_{n'}G_\phi + \eta \left(\frac{1}{2}\partial_{\tau'}G_S - \partial_{\tau'}G_\phi \right) \star K_H$$

⋮

Discretization

- High order triangulation of Ω and panelization of $\partial\Omega$
- Quadrature/interpolation via *fmm3dbie* and *chunkLE*
- Standard methods, need log singular quadratures on triangles
- FFT for acceleration of far-field interactions (non-PDE kernels)
- MATLAB as glue



Acceleration cartoon

Translational invariant kernel

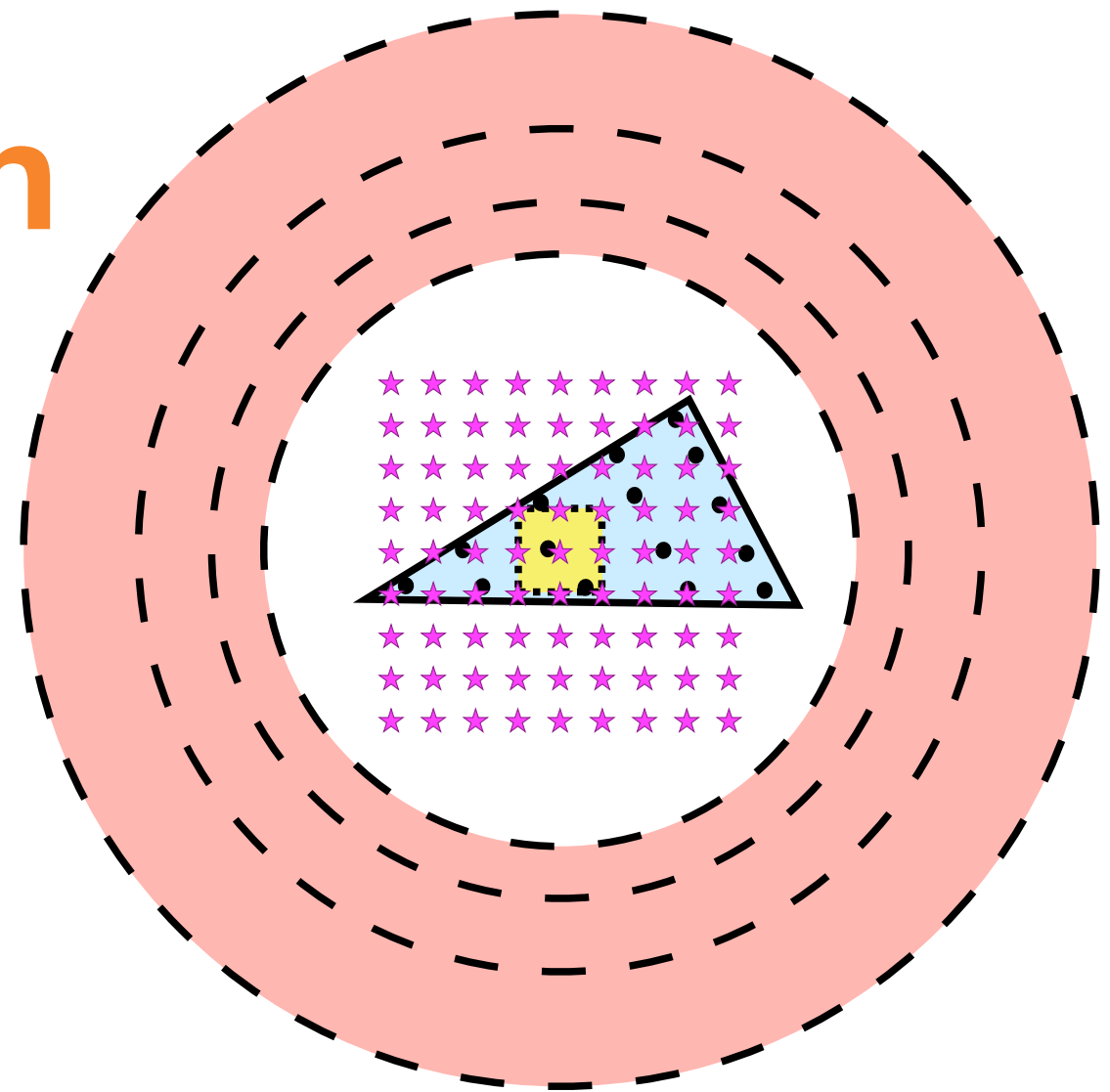
$K(x - y)$ + rectangular grid

$z_1, \dots, z_{N^2} \implies$ Fast apply via FFT

Our grid isn't rectangular! :(

Basic algorithm

- Spread to rectangular grid
- Use FFT to apply 2D convolution on grid
- Gather from rectangular grid
- Correct locally (need to do quadrature anyway)



For each y in yellow bin, solve

$$\min_{\alpha} \sum_{x_i \in \text{rings}} \left| \sum_{z_j \in \text{stars}} K(x_i - z_j) \alpha_j(y) - K(x_i - y) \right|^2$$

Same system on every bin!

Acceleration cartoon

Translational invariant kernel

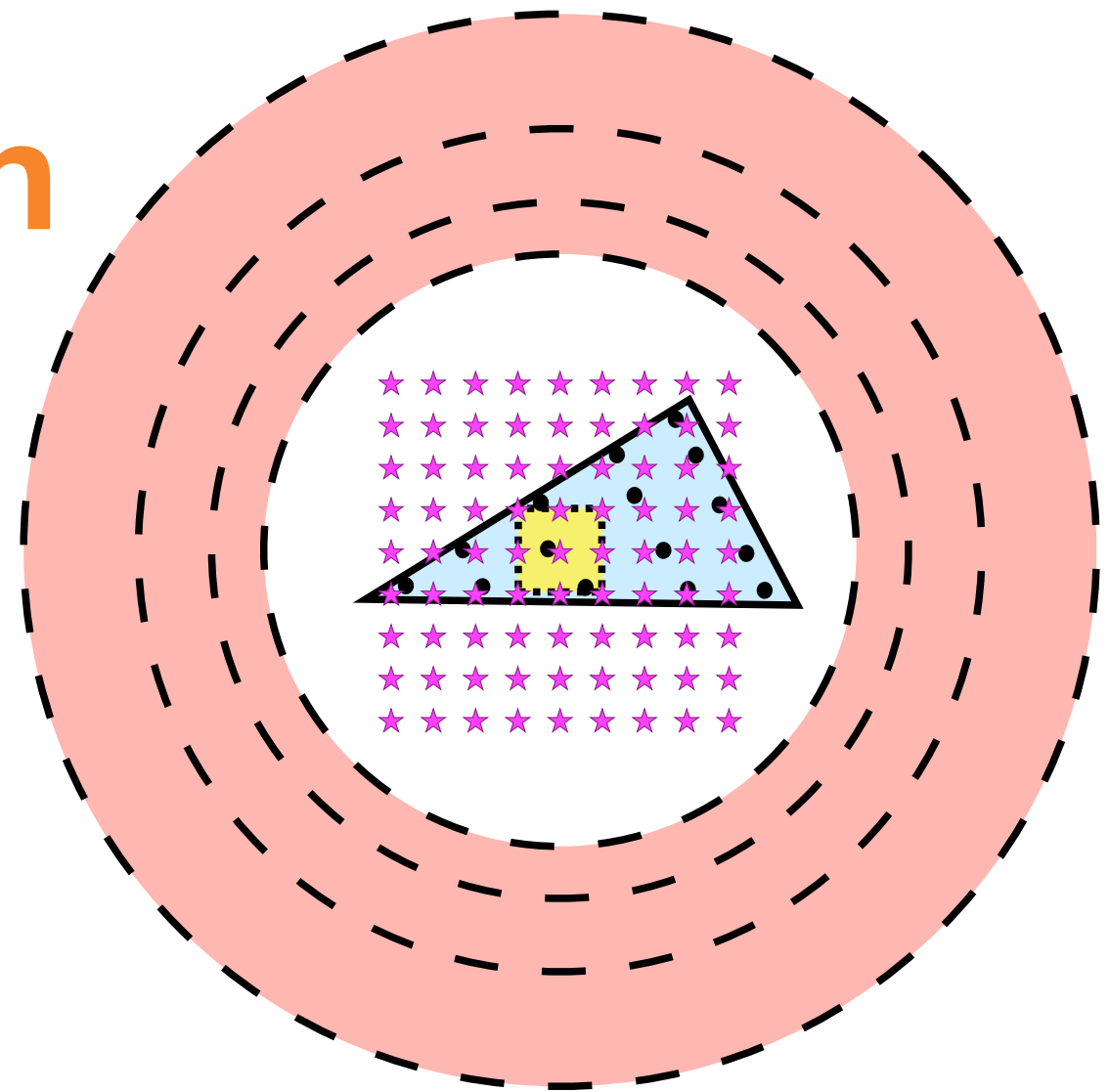
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Acceleration cartoon

Translational invariant kernel

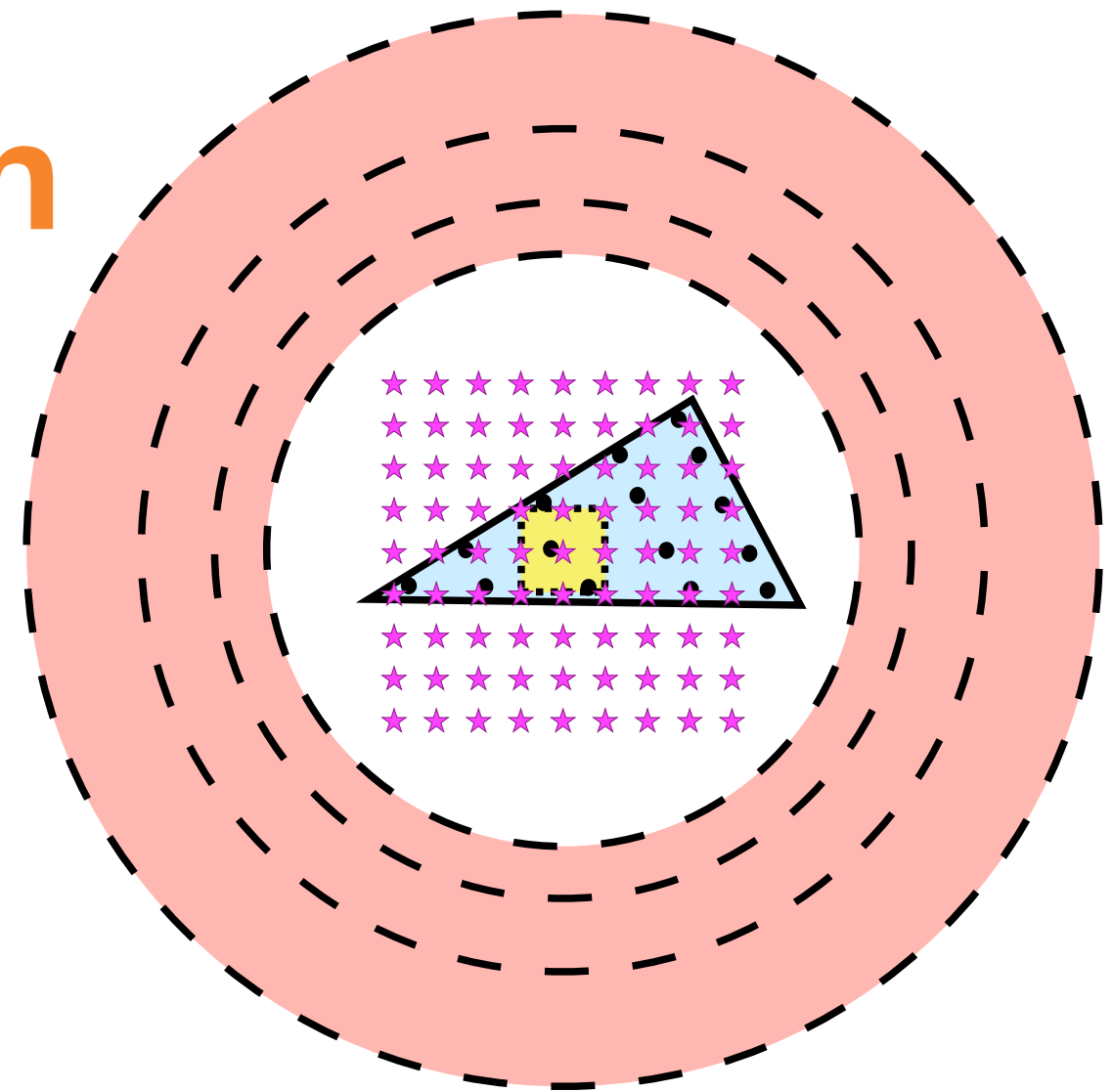
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Our grid isn't rectangular! :(

Basic algorithm

- Spread to rectangular grid
- Use FFT to apply 2D convolution on grid
- Gather from rectangular grid
- Correct locally (need to do quadrature anyway)



For each x in yellow bin, solve

$$\min_{\alpha} \sum_{y_i \in \text{rings}} \left| \sum_{z_j \in \text{stars}} K(z_j - y_i) \beta_j(x) - K(x - y_i) \right|^2$$

Same system on every bin!

Acceleration cartoon

Translational invariant kernel

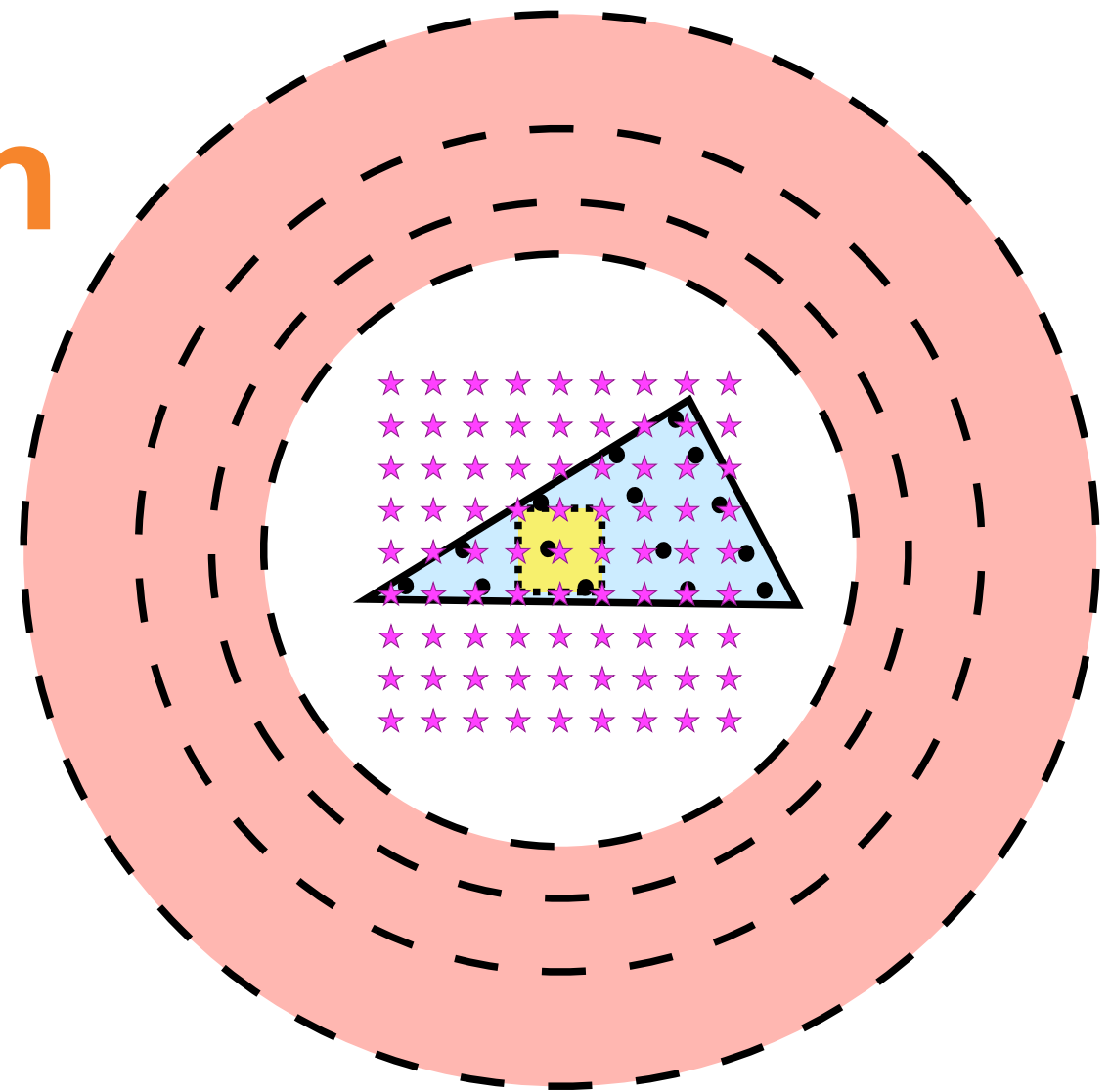
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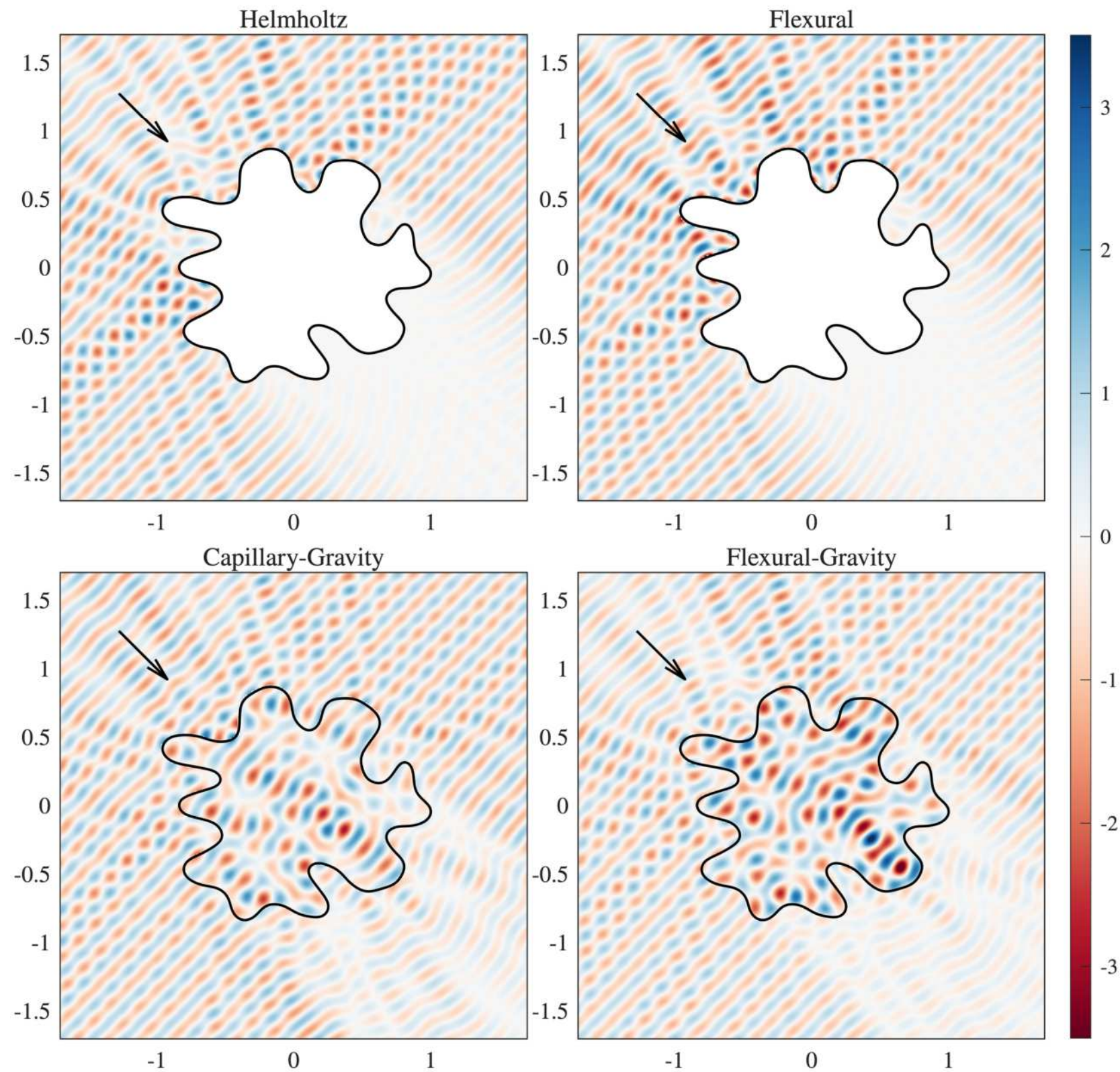
Basic algorithm

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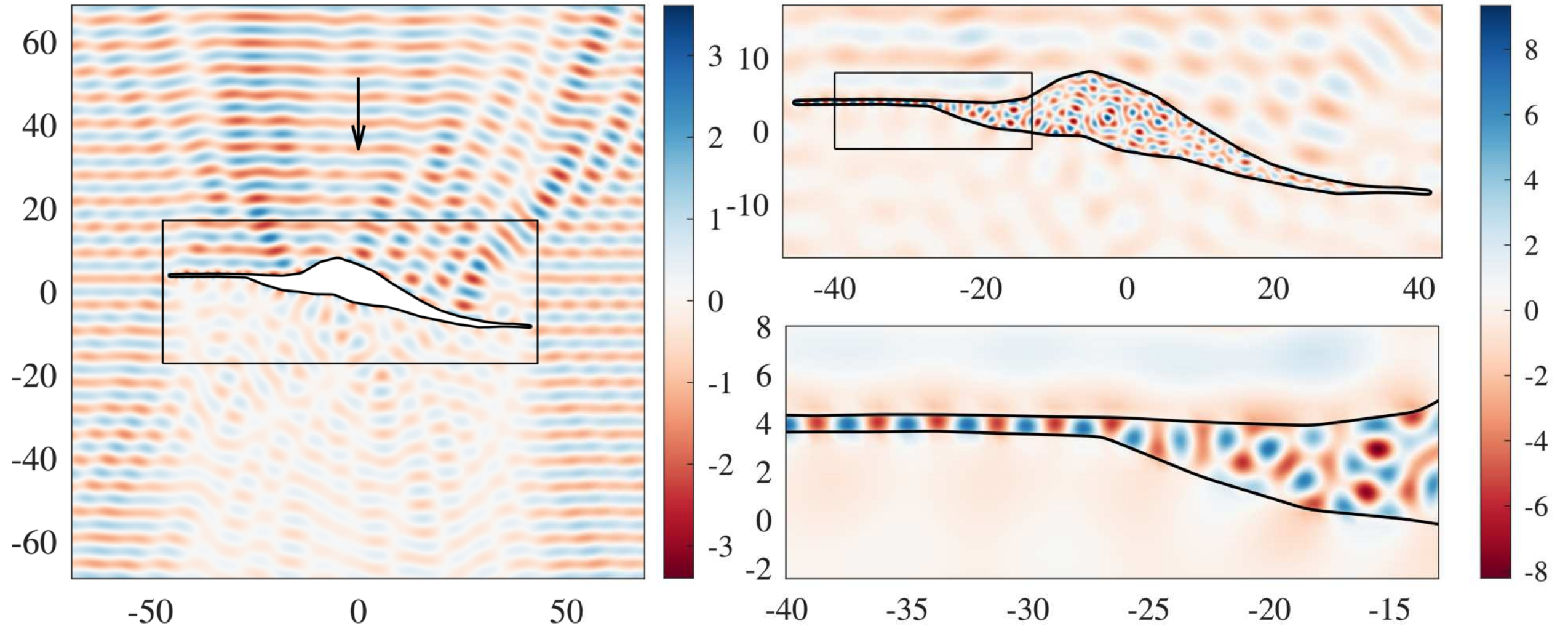


For all dots within the first ring, add correct contribution and subtract spreading/gathering contribution.

Scattering comparison



WR2 rift in Ross Ice Shelf



Conclusion/future directions

- A class of interface problems that reduces to second kind integral equations
- Explicit formulae for Green's functions
- Can adopt layer potential representations from related PDEs
- Standard numerical tools can be applied; kernel compression via proxy annuli
- P. Nekrasov is pursuing applications in geosciences
- Extensions to general curved surfaces

Thanks for listening!

Questions?